

Survey in Geometry

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1 Review of differentiable manifolds

Definition 1.1 (n-dimensional differentiable manifold). An *n-dimensional differentiable manifold* is a topological space M with an open cover $\{U_\alpha\}_\alpha$ and maps

$$\{\phi_\alpha : U_\alpha \rightarrow \phi_\alpha(U_\alpha) \subset \mathbb{R}^n\}_{\alpha \in \Lambda}$$

satisfying

- (i) $\phi_\alpha(U_\alpha) \subset \mathbb{R}^n$ is open and $\phi_\alpha : U_\alpha \rightarrow \phi_\alpha(U_\alpha)$ is a homeomorphism.
 - (ii) If $U_\alpha \cap U_\beta \neq \emptyset$, then the transition map $\phi_\beta \circ \phi_\alpha^{-1}$ is a diffeomorphism (between open subsets of $\mathbb{R}^n$).
- $\{(U_\alpha, \phi_\alpha) : \alpha \in \Lambda\}$ is called a *coordinate neighborhood system* or *atlas*. ■

Some terminology

- $n = \dim M$ *dimension* of M .
- ϕ_α *coordinate map* or *chart*.
- $\phi_\alpha = (x^1, \dots, x^n)$ *local coordinates* $x^i : U_\alpha \rightarrow \mathbb{R}$.

Definition 1.2 (Compatible charts). Assume M is an n-differentiable manifold with atlas $\{(U_\alpha, \phi_\alpha) : \alpha \in \Lambda\}$. A chart (U, ϕ) is *compatible* with the atlas if $\{(U_\alpha, \phi_\alpha)\}_{\alpha \in \Lambda} \cup (U, \phi)$ is again an atlas. The set of all such charts is called the *maximal coordinate neighborhood system* (or *atlas*) of M . ■

Henceforth, we assume the atlases of M are always maximal.

1.1 Differential of maps and tangent spaces

Definition 1.3 (C^∞). Assume M, N are differentiable manifolds. A map continuous $f : M \rightarrow N$ is C^∞ at $p \in M$ if there exists charts (U, ϕ) around p and (V, ψ) around $f(p)$ such that $\psi \circ f \circ \phi^{-1} : \phi(U) \rightarrow \psi(V)$ is C^∞ . ■

Remark 1.4. • $M = I \subset \mathbb{R}, c : I \rightarrow N$ is a C^∞ curve.

- $N = \mathbb{R}, f : M \rightarrow \mathbb{R}$ is a C^∞ function.
- $C^\infty(M) := \{C^\infty \text{ function on } M\}$. ■

Definition 1.5 (Tangent vector and tangent space). A map $v : C^\infty(M) \rightarrow \mathbb{R}$ is a *tangent vector* at $p \in M$ if for all $a, b \in \mathbb{R}$ and all $f, g \in C^\infty(M)$, we have

- (i) $v(af + bg) = av(f) + bv(g)$ (linearity).
- (ii) $v(fg) = f(p)v(g) + g(p)v(f)$ (Leibniz rule).

The set $T_p M := \{v : v \text{ is a tangent vector at } p\}$ is the *tangent space* to M at p . ■

Definition 1.6 (Velocity vector). Let $c : I \rightarrow M$ be a C^∞ curve with $c(0) = p \in M$. The *velocity vector* of c at p is the map $c'(0) \in T_p M$ defined by

$$c'(0)(f) = \left. \frac{d}{dt} \right|_{t=0} (f \circ c)(t), \quad f \in C^\infty(M).$$

■

Definition 1.7 (Coordinate vector fields). Let (U, ϕ) be a chart around p with $\phi(p) = 0$, $\phi = (x^1, \dots, x^n)$. Define curves

$$c_i(t) := \phi^{-1}(0, \dots, t, \dots, 0).$$

Then

$$\left. \frac{\partial}{\partial x^i} \right|_p := c'_i(0).$$

■

Theorem 1.8. For $p \in M$ an n -differentiable manifold, the tangent space $T_p M$ is an n -dimensional vector space with basis $\left\{ \left. \frac{\partial}{\partial x^1} \right|_p, \dots, \left. \frac{\partial}{\partial x^n} \right|_p \right\}$. I.e.

$$T_p M = \text{span} \left\{ \left. \frac{\partial}{\partial x^1} \right|_p, \dots, \left. \frac{\partial}{\partial x^n} \right|_p \right\}.$$

Definition 1.9 (Differential). Let $f : M \rightarrow N$ be a C^∞ map between differentiable manifolds. The *differential* of f at $p \in M$ is the map $df_p : T_p M \rightarrow T_{f(p)} N$ defined by

$$df_p(v)(g) := (f \circ c)'(0), \quad c \in C^\infty, \quad c(0) = p \text{ and } c'(0) = v.$$

■

Remark 1.10. c in the above definition is not unique, but $df_p(v)$ is well-defined.

■

Definition 1.11 (Diffeomorphism). Let M, N be differentiable manifolds. $f : M \rightarrow N$ is a *diffeomorphism* if

- f is a bijection.
- f and f^{-1} are C^∞ .

In this case, M and N are *diffeomorphic*.

■

1.2 Diffeomorphism groups and quotient manifolds

Let M be an n -differentiable manifold.

Definition 1.12 (C^∞ transformation group). A set of diffeomorphisms $\Gamma = \{\gamma_i\}_{i \in I}$ on M is a C^∞ transformation group if

- For all $\gamma_i, \gamma_j \in \Gamma$, we have $\gamma_i \circ \gamma_j \in \Gamma$.
- For all $\gamma_i \in \Gamma$, we have $\gamma_i^{-1} \in \Gamma$.
- The identity map Id_M is in Γ .

■

Definition 1.13 (Proper discontinuous transformation group). A C^∞ transformation group Γ on M is *properly discontinuous* if for each $p \in M$, there exists a chart (U, ϕ) around p such that

$$\gamma_i \in \Gamma \setminus \{\text{Id}_M\} \implies \gamma_i(U) \cap U = \emptyset \text{ for all } i \in I.$$

■

Let Γ be a properly discontinuous transformation group on M . For each $p \in M$, let (U_p, ϕ_p) be a chart around p as in definition 1.13.

Lemma 1.14. *For all $p \in M$ and all $\gamma_i, \gamma_j \in \Gamma$ ($\gamma_i \neq \gamma_j$), we have $\gamma_i(U_p) \cap \gamma_j(U_p) = \emptyset$ or $\gamma_i(U_p) = \gamma_j(U_p)$.*

Proof. $\gamma_i \neq \gamma_j \implies \gamma_i^{-1} \circ \gamma_j \neq \text{Id}_M$. Then, by definition 1.13, $U_p \cap \gamma_i^{-1} \circ \gamma_j(U_p) = \emptyset \implies \gamma_i(U_p) \cap \gamma_j(U_p) = \emptyset$. $\square$

Definition 1.15 (Orbit equivalence relation). Let Γ be a group acting on M . Define a relation on M by

$$p \sim q \iff \exists \gamma \in \Gamma \text{ such that } q = \gamma(p).$$

■

Proposition 1.16. *The relation $\sim$ is an equivalence relation.*

Definition 1.17 (Orbit). The equivalence class of p ,

$$[p] = \{\gamma(p) \mid \gamma \in \Gamma\} \in M / \sim,$$

is the *orbit* of p . $\square$

Definition 1.18 (Quotient manifold). The quotient set $M/\Gamma := M / \sim$ is called the *quotient manifold* of M by Γ .

- M/Γ is equipped with the quotient topology via the projection map $\pi : M \rightarrow M/\Gamma$.
- $D \subset M$ is a *fundamental domain* if $\pi|_D : D \rightarrow M/\Gamma$ is surjective. $\square$

To show that M/Γ is a differentiable manifold, we construct charts. For $p \in M$, let $\hat{U}_p := \pi(U_p) \subset M/\Gamma$.

Lemma 1.19. *If $q, r \in U_p$ satisfy $q \sim r$ then $q = r$. In particular, $\pi|_{U_p} : U_p \rightarrow \hat{U}_p$ is a bijection.*

Proof. $q \sim r \implies \exists \gamma_i \in \Gamma : r = \gamma_i(q) \in \gamma_i(U_p)$. Thus $r \in \gamma_i(U_p)$ for some $\gamma_i \in \Gamma$. Since Γ is properly discontinuous, $\gamma_i = \text{Id}_M$. Hence, $r = \gamma_i(q) = q$. The second claim follows from $\pi(q) = \pi(r) \iff q \sim r$. $\square$

Problem 1. Define $\hat{\phi} : \hat{U}_p \rightarrow \hat{\phi}_p(\hat{U}_p) = \phi_p(U_p) \subset \mathbb{R}^n$ by

$$\hat{\phi}_p := \phi_p \circ (\pi|_{U_p})^{-1}.$$

Show that $\hat{\phi}_p$ is a homeomorphism. (Hint: In general, quotient maps are not necessarily open; use the properly discontinuous property). $\square$

Theorem 1.20. $\{(\hat{U}_p, \hat{\phi}_p) : p \in M\}$ is an atlas for M/Γ . Hence, M/Γ is a differentiable manifold.

1.3 Topology of differentiable manifolds

Recall the convergence of a sequence $\{p_i\}_{i=1}^\infty$:

$$\lim_{i \rightarrow \infty} p_i = p \iff \forall \text{ open } O \ni p, \exists i_0 \in \mathbb{N} : i > i_0 \implies p_i \in O.$$

Proposition 1.21 (Convergence in differentiable manifolds). $\lim_{i \rightarrow \infty} p_i = p$ if and only if for every chart (U, ϕ) around p , we have $\lim_{i \rightarrow \infty} \phi(p_i) = \phi(p)$ (in $\mathbb{R}^n$). That is,

$$\lim_{i \rightarrow \infty} p_i = p \iff \forall (U, \phi) \exists i_0 \in \mathbb{N} : i > i_0 \implies \phi(p_i) \rightarrow \phi(p).$$

Remark 1.22. M is Hausdorff and second countable and locally Euclidean. Thus, $f : M \rightarrow N$ is continuous at $p \in M$ if and only if $p_i \rightarrow p$ implies $f(p_i) \rightarrow f(p)$. ■

Lemma 1.23. Let (U, ϕ) be a chart, $c : (a, b) \rightarrow M$ be a C^∞ curve and $t_0 \in (a, b)$ that satisfy

- (i) The map $\phi \circ c : (a, b) \rightarrow \mathbb{R}^n$ is C^∞ .
- (ii) $\exists \{t_i\}_{i=1}^\infty \subset (a, b) : t_i \rightarrow t_0$ and $t_i \geq t_0$ and $c(t_i) \notin U$.

Then

- (i) $c(t_0) \notin U$.
- (ii) $\forall \{s_i\}_{i=1}^\infty \subset (a, b) : s_i \rightarrow t_0$ and $s_i < t_0$, $(\phi \circ c)(s_i)$ does not converge in $\phi(U)$.

Proof. (1) Assume $c(t_0) \in U$. By continuity, $\exists \delta_0 > 0$ such that $c((t_0 - \delta_0, t_0 + \delta_0)) \subset U$. For sufficiently large i , we have $t_i \in (t_0, t_0 + \delta_0) \implies c(t_i) \in U$, which contradicts (ii). (2) Assume $\exists \{s_i\} \rightarrow t_0$ ($s_i < t_0$) such that $\{\phi \circ c(s_i)\}$ converges in $\phi(U)$. Then $(\phi \circ c)(s_i) \rightarrow x_0$ for some $x_0 \in \phi(U)$. By continuity of ϕ^{-1} , $c(s_i) = (\phi^{-1} \circ \phi)(c(s_i)) \rightarrow \phi^{-1}(x_0)$ as $i \rightarrow \infty$. By the continuity of c , we have $c(s_i) \rightarrow c(t_0)$ as $i \rightarrow \infty$. Since M is Hausdorff, the limit is unique, so $\phi^{-1}(x_0) = c(t_0)$. Thus, $c(t_0) = \phi^{-1}(x_0) \in U$, which contradicts (1). □

1.4 Tangent bundle

Let M be a differentiable manifold.

Definition 1.24 (Tangent bundle). The *tangent bundle* of M is the set

$$TM := \bigsqcup_{p \in M} T_p M = \{(p, v) : p \in M, v \in T_p M\}.$$

■

We introduce a differentiable structure on TM using the following framework:

Problem 2. Let $\{X_\lambda, \mathcal{O}_\lambda\}_{\lambda \in \Lambda}$ be a family of topological spaces satisfying the *gluing axiom*:

$$X_\lambda \cap X_\mu \neq \emptyset \implies \begin{array}{l} \text{the relative topologies from } X_\lambda \text{ and } X_\mu \text{ coincide on } X_\lambda \cap X_\mu, \\ \text{and } X_\lambda \cap X_\mu = \mathcal{O}_\lambda \cap \mathcal{O}_\mu. \end{array} \quad (1.1)$$

Set $X := \bigsqcup_{\lambda \in \Lambda} X_\lambda$.

- (1) Show that $\mathcal{O} := \{U \subset X : X_\lambda \cap U \in \mathcal{O}_\lambda \forall \lambda \in \Lambda\}$ is a topology on X .
- (2) Show that the relative topology on X_λ from $(X, \mathcal{O})$ coincides with $\mathcal{O}_\lambda$ for each $\lambda \in \Lambda$. ■

Remark 1.25. The gluing axiom 1.1 is necessary for (2) and to ensure $X_\lambda \in \mathcal{O}$. Such a topology $\mathcal{O}$ is uniquely determined by the base $\bigcup_{\lambda \in \Lambda} \mathcal{O}_\lambda$. ■

Let (U, ϕ) be any chart on M and set

$$TU := \bigsqcup_{p \in U} T_p M = \{(p, v) : p \in U, v \in T_p M\} \subset TM.$$

Define $\tilde{\phi} : TU \rightarrow \phi(U) \times \mathbb{R}^n \subset \mathbb{R}^{2n}$ as follows: For $p \in U$, $v \in T_p M$, let $\phi(p) = (x_1, \dots, x_n)$ and $v = \sum_{i=1}^n v_i \partial_i$, then

$$\tilde{\phi}(p, v) := (x_1, \dots, x_n, v_1, \dots, v_n).$$

Theorem 1.26. (1) The family of topological spaces $\{TU\}$ satisfies the gluing axiom 1.1. Thus, $TM = \bigsqcup_{(U, \phi)} TU$ is a topological space with the topology defined in problem 2.

- (2) TM is a $2n$ -dimensional manifold with atlas $\{(TU, \tilde{\phi})\}$.
- (3) The projection $\pi : TM \rightarrow M$ defined by $\pi(p, v) := p$ for $v \in T_p M$ ($p \in M$) is a C^∞ map.

2 Tensors and Tensor Fields

2.1 Tensor Algebra

We generalize bilinear forms to multilinear forms.

Definition 2.1 (Multilinear map and multilinear function). Let $V_1, \dots, V_k$ and W be real vector spaces. A map

$$f : V_1 \times \dots \times V_k \rightarrow W$$

is *multilinear* if it is linear in each component. For example,

$$f(av_1 + bw_1, v_2, \dots, v_k) = af(v_1, v_2, \dots, v_k) + bf(w_1, v_2, \dots, v_k)$$

for all $a, b \in \mathbb{R}$ and $v_1, w_1 \in V_1$. If $W = \mathbb{R}$, f is called a *multilinear function*. ■

Definition 2.2 ((s, t) -type tensor). Let V be an n -dimensional real vector space and V^* its dual.

For $s, t \in \mathbb{Z}_{\geq 0}$, a multilinear map

$$T : (V^*)^s \times V^t \rightarrow \mathbb{R},$$

is called a *tensor of type* (s, t) . The set of all such tensors is denoted by $T_t^s(V)$; it is a real vector space under pointwise addition and scalar multiplication. ■

Remark 2.3. Since the evaluation map $V \rightarrow V^{**}$ $v^{**}(f) := f(v)$ is a linear isomorphism, it is sufficient to consider only V^* and V in definition 2.1 ■

Problem 3. For a multilinear map $T : (V^*)^s \times V^t \rightarrow \mathbb{R}$, define $\hat{T} \in T_t^{s+1}(V)$ by

$$\hat{T}(\omega^1, \dots, \omega^{s+1}, v_1, \dots, v_t) := \omega^{s+1}(T(\omega^1, \dots, \omega^s, v_1, \dots, v_t)),$$

where $\omega^1, \dots, \omega^{s+1} \in V^*$ and $v_1, \dots, v_t \in V$. Show that the map $T \mapsto \hat{T}$ is a linear isomorphism. ■

Definition 2.4 (Tensor product). For $T \in T_t^s(V)$ and $S \in T_{t'}^{s'}(V)$, define $T \otimes S \in T_{t+t'}^{s+s'}(V)$ by

$$\begin{aligned} (T \otimes S)(\omega^1, \dots, \omega^{s+s'}, v_1, \dots, v_{t+t'}) \\ := T(\omega^1, \dots, \omega^s, v_1, \dots, v_t) \times S(\omega^{s+1}, \dots, \omega^{s+s'}, v_{t+1}, \dots, v_{t+t'}), \end{aligned}$$

where $\omega^1, \dots, \omega^{s+s'} \in V^*$ and $v_1, \dots, v_{t+t'} \in V$. ■

Remark 2.5. For any tensors T, S, R and $a \in \mathbb{R}$, we have:

- (i) $(T \otimes S) \otimes R = T \otimes (S \otimes R)$.
- (ii) $(aT) \otimes S = T \otimes (aS) = a(T \otimes S)$.

These are denoted as $T \otimes S \otimes R$ and $aT \otimes S$, respectively. Furthermore, S and R have the same type, the distributive laws hold :

- (i) $T \otimes (S + R) = T \otimes S + T \otimes R$.
- (ii) $(S + R) \otimes T = S \otimes T + R \otimes T$.

Remark 2.6. In general, $S \otimes T \neq T \otimes S$. ■

Let $\{e_i\}$ be a basis for V and $\{e^i\}$ be its dual. For $T \in T_t^s(V)$, the n^{s+t} real numbers

$$T_{j_1 \dots j_t}^{i_1 \dots i_s} := T(e^{i_1}, \dots, e^{i_s}, e_{j_1}, \dots, e_{j_t}),$$

are called the *coefficients* (or *components*) of T . For $T \in T_t^s(V)$ and $S \in T_{t'}^{s'}(V)$, the coefficients of $T \otimes S$ are given by

$$(T \otimes S)_{j_1 \dots j_{t+t'}}^{i_1 \dots i_{s+s'}} = T_{j_1 \dots j_t}^{i_1 \dots i_s} S_{j_{t+1} \dots j_{t+t'}}^{i_{s+1} \dots i_{s+s'}}.$$

Problem 4. Show that the set of n^{s+t} tensors

$$e_{i_1} \otimes \dots \otimes e_{i_s} \otimes e^{j_1} \otimes \dots \otimes e^{j_t},$$

where $i_1, \dots, i_s, j_1, \dots, j_t = 1, \dots, n$ forms a basis for $T_t^s(V)$; and that for any $T \in T_t^s(V)$,

$$T = \sum_{i_1, \dots, i_s, j_1, \dots, j_t} T_{j_1 \dots j_t}^{i_1 \dots i_s} e_{i_1} \otimes \dots \otimes e_{i_s} \otimes e^{j_1} \otimes \dots \otimes e^{j_t}.$$

In particular, show that $\dim T_t^s(V) = n^{s+t}$. We often write

$$T_t^s(V) = V \otimes \dots \otimes V \otimes V^* \otimes \dots \otimes V^*.$$

■

2.2 Exterior Algebra

Definition 2.7 (p-form). Let V be a real vector space and $p \in \mathbb{N}$. An element of $T_p^0(V)$ is called a *p-form*. ■

Remark 2.8. A p-form ω is *skew symmetric*. That is,

$$\omega(v_{\sigma(1)}, \dots, v_{\sigma(p)}) = \text{sgn}(\sigma) \omega(v_1, \dots, v_p),$$

for any $v_1, \dots, v_p \in V$ and $\sigma \in S_p$. Furthermore, $\Lambda^p V^* := \{\text{skew symmetric p-forms}\}$ is a vector space. ■

Definition 2.9 (Exterior product). For $\alpha \in \Lambda^p(V^*)$ and $\beta \in \Lambda^q(V^*)$, the *exterior product* $\alpha \wedge \beta$ is defined by

$$\alpha \wedge \beta := \frac{1}{p!q!} \sum_{\sigma \in S_{p+q}} \text{sgn}(\sigma) \alpha \otimes \beta (v_{\sigma(1)}, \dots, v_{\sigma(p+q)}).$$

■

Proposition 2.10. Let $\alpha, \alpha^1, \alpha^2 \in \Lambda^p(V^*)$ and $\beta, \beta^1, \beta^2 \in \Lambda^q(V^*)$.

- (i) $(a\alpha^1 + b\alpha^2) \wedge \beta = a\alpha^1 \wedge \beta + b\alpha^2 \wedge \beta$.
- (ii) $\alpha \wedge (a\beta^1 + b\beta^2) = a\alpha \wedge \beta^1 + b\alpha \wedge \beta^2$.
- (iii) $(\alpha \wedge \beta) \wedge \gamma = \alpha \wedge (\beta \wedge \gamma)$.
- (iv) $\beta \wedge \alpha = (-1)^{pq} \alpha \wedge \beta$.

2.3 Tensor Fields and differential forms on manifolds

Let M be an n -dimensional manifold and (V, ϕ) a chart with $\phi = (x^1, \dots, x^n)$. At each $p \in U$, applying Proposition 2.10 with $V = T_p M$, yields

$$\frac{\partial}{\partial x^{i_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{i_s}} \otimes dx^{j_1} \otimes \dots \otimes dx^{j_t},$$

which forms a basis for $T_t^s(T_p M)$. Any $T_p \in T_t^s(T_p M)$ can be written as,

$$T_p = \sum_{i_1, \dots, i_s, j_1, \dots, j_t} T_{j_1 \dots j_t}^{i_1 \dots i_s} \frac{\partial}{\partial x^{i_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{i_s}} \otimes dx^{j_1} \otimes \dots \otimes dx^{j_t}.$$

Definition 2.11 ((s, t) -type tensor field). A map $T : M \ni p \rightarrow T_p \in T_t^s(T_p M)$ is a (s, t) -type tensor field. T is C^∞ if all its coefficients $T_{j_1 \dots j_t}^{i_1 \dots i_s}$ are C^∞ functions on any chart (V, ϕ) . ■

Remark 2.12. It suffices to check the C^∞ property on a single atlas. ■

3 Riemannian metric

Assume that M is Hausdorff differentiable manifold.

Definition 3.1 (Riemannian metric and Riemannian manifold). A *Riemannian metric* g is a map

$$M \ni p \rightarrow \langle \cdot, \cdot \rangle_p =: g_p(\cdot, \cdot) \text{ that assigns a,}$$

(positive definite) inner product to each $T_p M$. The pair (M, g) is called a *Riemannian manifold*. In terms of tensor field, g is a $(0, 2)$ -type tensor field (or a 2-form) on M . ■

Definition 3.2 (Length and angle). For $v, w \in T_p M$, we have the following notions of length and angle:

- *Length* $\|v\| := \sqrt{\langle v, v \rangle}$.
- *Angle* : $\theta \in [0, \pi]$ such that $\cos \theta = \frac{\langle v, w \rangle}{\|v\| \|w\|}$.

■

Example 3.3. Let (v, ϕ) be a chart with

$$\phi = (x^1, \dots, x^n).$$

Define $g_{ij} : U \rightarrow \mathbb{R}$ by

$$g_{ij}(p) := \left\langle \left(\frac{\partial}{\partial x^i} \right)_p, \left(\frac{\partial}{\partial x^j} \right)_p \right\rangle.$$

On U , g is expressed as

$$g = \sum_{i,j=1}^n g_{ij} dx^i \otimes dx^j.$$

The matrix (g_{ij}) is a positive definite symmetric matrix.

$$ds^2 = \sum_{i,j=1}^n \delta_{ij} dx^i \otimes dx^j = \sum_{i=1}^n (dx^i)^2.$$

■

Problem 5. Consider two charts (U, ϕ) and (V, ψ) with $U \cap V \neq \emptyset$.

$$\phi = (x^1, \dots, x^n), \psi = (y^1, \dots, y^n),$$

and let g_{ij} , $\hat{g}_{kl}$ be the components of g with respect to these charts. Show that on $U \cap V$ satisfies the following transformation law:

$$\hat{g}_{kl} = \sum_{i,j=1}^n g_{ij} \frac{\partial x^i}{\partial y^k} \frac{\partial x^j}{\partial y^l}.$$

■

Definition 3.4 (C^∞ Riemannian manifold). (M, g) is a C^∞ Riemannian manifold if g is a C^∞ tensor field. ■

Proposition 3.5. The following are equivalent:

- (M, g) is a C^∞ Riemannian manifold.
- Components g_{ij} are C^∞ functions for any chart (U, ϕ) .
- $p \rightarrow \langle X_p, Y_p \rangle$ is C^∞ for any C^∞ vector fields X, Y .

Henceforth, assume that all Riemannian metrics are C^∞ .

Remark 3.6. g, h : Riemannian metric, $f \in c^\infty(M, \mathbb{R}_{>0}) \implies fg$ and $g + h$ are also Riemannian metrics. ■

Proposition 3.7. Every paracompact manifold admits a Riemannian metric.

Proof. Let $\{(U_k, \phi_k)\}_{k=1}^{K_0}$ be a finite atlas and let $\{x_k\}_{k=1}^{K_0}$ be a partition of unity subordinate to U_k . On U_k , define g_k

$$g_k(u, w) := \langle (d\phi_k)_p(v), (d\phi_k)_p(w) \rangle_{\mathbb{R}^n}, v, w \in T_p M.$$

Then the global metric is given by:

$$g := \sum_{k=1}^{K_0} x_k g_k$$

□

3.1 Isometries

Definition 3.8 (Isometry). Let (M, g) and (N, h) be Riemannian manifolds and let $f : M \rightarrow N$ be a smooth map.

- The map f is called *locally isometric* if

$$\langle (df)_p(v), (df)_p(w) \rangle_{f(p)} = \langle v, w \rangle_p \quad \text{for all } p \in M, v, w \in T_p M.$$

- The map f is called an *isometry* if it is a diffeomorphism and locally isometric.
- The manifolds M and N are called *isometric* if there exists an isometry $f : M \rightarrow N$.

■

Proposition 3.9. (1) $f : L \rightarrow M, h : M \rightarrow N$ are (local) isometric maps $\implies h \circ f$ is a (local) isometric map.

(2) f is isometry $\implies f^{-1}$ is isometry.

Thus, the relation "Isometric" is an equivalence relation.

3.2 Riemannian submanifold

Definition 3.10 (Riemannian submanifold). Assume M is a Riemannian manifold and $N \subset M$ is a submanifold. N is a *Riemannian submanifold* if the metric is defined by $\langle v, w \rangle_N := \langle v, w \rangle_M$ for all $v, w \in T_p N$. ■

The inclusion $C : N \rightarrow M$ is always an isometric map.

3.2.1 Remember 3.9 Nash Embedding Theorem

Every Riemannian manifold is isometric to a Riemannian submanifold of some Euclidian space $\mathbb{R}^n$. $\Leftrightarrow$ *Whitney Embedding Theorem*

3.3 Riemannian product and warped product

Definition 3.11 (Riemannian product and product Riemannian manifold). Let M, N be Riemannian manifolds. The *Riemannian product* $M \times N$ is defined by

$$T_{(p,q)}(M \times N) \simeq T_p M \times T_q N, \quad (p, q) \in M \times N.$$

The metric is defined by

$$\langle v, w \rangle_{M \times N} := \langle v_1, w_1 \rangle + \langle v_2, w_2 \rangle, \quad v = (v_1, v_2), w = (w_1, w_2) \in T_p M \times T_q N.$$

We call $(M \times N, \langle \cdot, \cdot \rangle_{M \times N})$ the *product Riemannian manifold* (of M and N).

In this metric, $T_p M$ and $T_q N$ are orthonormal to each other and each factor remains its original inner product. ■

Example 3.12. The Riemannian product $\mathbb{R}^m \times \mathbb{R}^n$ is isometric to $\mathbb{R}^{m+n}$. ■

Definition 3.13 (Warped product). Let M, N be Riemannian manifolds and let $f : M \rightarrow \mathbb{R}_{>0}$ be a C^∞ function. The *warped product* $M \times_f N$ is the product manifold $M \times N$ equipped with the metric

$$\langle v, w \rangle_{M \times_f N} := \langle v_1, w_1 \rangle + f(p)^2 \langle v_2, w_2 \rangle, \quad v = (v_1, v_2), w = (w_1, w_2) \in T_{(p,q)}(M \times N).$$

We call the function f the *warping function*. ■

Problem 6. Let $(0, +\infty)$ have the Euclidean metric and $\mathbb{S}^{n-1} \subset \mathbb{R}^n$ have the standard metric. Show that the warped product $(0, +\infty) \times_r \mathbb{S}^{n-1}$ is isometric to $(\mathbb{R}^n \setminus \{0\}, g_{\mathbb{R}^n})$, where $r(x) := x$, $x \in (0, +\infty)$ is the warping function. ■

3.4 Riemannian quotient manifolds

Definition 3.14 (C^∞ transformation group). A C^∞ *transformation group* $P = \{\gamma_i\}_{i \in I}$ on a Riemannian manifold M is a group of isometries if each $\gamma_i \in P$ is an isometry. ■

Let M be a Riemannian manifold and P be a properly discontinuous group of isometries on M .

Lemma 3.15. Suppose $p, q \in M$ satisfy $p \sim q$. For any $v, w \in T_{[p]}(M/\Gamma)$, we have

$$\langle d(\pi|_{U_p}^{-1})(v), d(\pi|_{U_p}^{-1})(w) \rangle = \langle d(\pi|_{U_q}^{-1})(v), d(\pi|_{U_q}^{-1})(w) \rangle.$$

I'm not sure what U represents here.

Proof. Since $p \sim q$, there exists $\gamma_i \in \Gamma$ such that $q = \gamma_i(p)$. Furthermore, since γ_i is an isometry and $\pi \odot \gamma_i = \pi$, it follows that

$$\begin{aligned} \langle d(\pi|_{U_q}^{-1})(v), d(\pi|_{U_q}^{-1})(w) \rangle &= \langle d\gamma_i(d\pi|_{U_p}^{-1}(v)), d\gamma_i(d\pi|_{U_p}^{-1}(w)) \rangle \\ &= \langle d\pi|_{U_p}^{-1}(v), d\pi|_{U_p}^{-1}(w) \rangle. \end{aligned}$$

□

We define a Riemannian metric on M/Γ by

$$\langle v, w \rangle_{M/\Gamma} := \langle d(\pi|_{U_p}^{-1})(v), d(\pi|_{U_p}^{-1})(w) \rangle, \quad v, w \in T_{[p]}(M/\Gamma).$$

With this metric, M/Γ is called a *quotient Riemannian manifold*. By construction, the projection $\pi : M \rightarrow M/\Gamma$ is a local isometry.

3.5 Volume element

Let M be an oriented n -dimensional Riemannian manifold.

Definition 3.16. For a positively oriented chart (U, ϕ) with $\phi = (x^1, \dots, x^n)$, the n -form on

$$\text{dvol} := \sqrt{|g|} dx^1 \wedge \dots \wedge dx^n, \quad |g| := \det(g_{ij}),$$

is called the *volume element* (or *volume form*) of M . ■

Remark 3.17. dvol is a symbol and not necessarily the exterior derivative or some $(n-1)$ -form. This form is globally well-defined on M . In fact, if $(U, \phi), (V, \psi)$ are two positively oriented charts on M , then on $U \cap V (\neq \emptyset)$,

$$\begin{aligned} \det(\hat{g}_k l) &= \det \left(\frac{\partial x^i}{\partial y^k} \right)^2 \det(g_{ij}) \\ \implies \sqrt{|\hat{g}|} &= \left| \det \left(\frac{\partial x}{\partial y} \right) \right| \sqrt{|g|} \text{ and} \\ dy^1 \wedge \dots \wedge dy^n &= \det \left(\frac{\partial y}{\partial x} \right) dx^1 \wedge \dots \wedge dx^n. \end{aligned}$$

Thus $\sqrt{|\hat{g}|} dy^1 \wedge \dots \wedge dy^n = \sqrt{|g|} dx^1 \wedge \dots \wedge dx^n$. ■

Definition 3.18 (Volume). The *volume* of an open subset $D \subset M$ is defined by:

$$\text{vol}(D) := \int_D \text{dvol}.$$

■

Remark 3.19. For non-orientable manifolds, the volume can still be defined by partitioning D into small orientable regions. ■

4 Length of curves and Riemann distance

Let M be a Riemannian manifold, $I \subset \mathbb{R}$ be an interval, and $c : I \rightarrow M$ a C^∞ curve.

Definition 4.1 (Length of a curve). For $a, b \in I$ such that $a < b$, the *length* $L(c|_{[a,b]})$ of c on $[a, b]$ is defined by

$$L(c|_{[a,b]}) := \int_a^b \|c'(t)\| dt.$$

Here, I believe $\|\dots\| = \sqrt{\langle \dots, \dots \rangle}$. ■

Definition 4.2 (Regular curve). A C^∞ curve $C : I \rightarrow M$ is *regular* if

$$c'(t) \neq 0, \quad \forall t \in I.$$

Theorem 4.3 (Inverse function theorem for regular curves). Assume that the parameter t is a C^∞ function of u on $J \subset \mathbb{R}$, i.e. $t = t(u)$ satisfying:

(i) $t : J \rightarrow I$ is surjective.

(ii) $\frac{dt}{du}(u) \neq 0 \forall u \in J$.

Then there exists a C^∞ inverse $t^{-1} : I \rightarrow J$. This is called a transformation of parameter t to u .

Let $\tilde{c} := c(t(u))$ be a reparametrization of c by t . Fix a chart $(U, \phi), \phi = (x^1, \dots, x^n)$ and subinterval $I' \subset I$ such that $c(I') \subset U$. For $t \in I'$ and $u \in t^{-1}(I')$, let

$$\phi(c(t)) = (c^1(t), \dots, c^n(t)), \quad \phi(\tilde{c}(u)) = (\tilde{c}^1(u), \dots, \tilde{c}^n(u)).$$

Since $\tilde{c}^i(u) = c^i(t(u))$, we have

$$\begin{aligned} \tilde{c}'(u) &= \sum_{i=1}^n \frac{d\tilde{c}^i}{du} \left(\frac{\partial}{\partial x^i} \right)_{\tilde{c}(u)} \\ &= \sum_{i=1}^n \frac{dt}{du} \frac{dc^i}{dt}(t(u)) \left(\frac{\partial}{\partial x^i} \right)_{\tilde{c}(u)} \\ &= \frac{dt}{du} c'(t(u)). \end{aligned}$$

Since (U, ϕ) was arbitrary, for all $u \in J$,

$$\tilde{c}'(u) = c'(t(u)) \frac{dt}{du}.$$

Proposition 4.4. Let $c : I \rightarrow M$ be a regular curve and $t = t(u)$ ($u \in J$) be a transformation. Then

- (1) If $\frac{dt}{du}(u) > 0$ ($\forall u \in J$), then $L(\tilde{c}|_{[a,b]}) = L(c|_{[t(a),t(b)]}) \forall a, b \in J (a < b)$.
(2) If $\frac{dt}{du}(u) < 0$ ($\forall u \in J$), then $L(\tilde{c}|_{[a,b]}) = L(c|_{[t(b),t(a)]}) \forall a, b \in J (a < b)$.

Proof. (1) By Definition 4.1 (length of a curve),

$$\begin{aligned} L(\tilde{c}|_{[a,b]}) &= \int_a^b \|\tilde{c}'(u)\| du \\ &= \int_a^b \|c'(t(u))\| \left| \frac{dt}{du} \right| du \\ &= \int_{t(a)}^{t(b)} \|c'(t)\| dt \\ &= L(c|_{[t(a),t(b)]}). \end{aligned}$$

(2) Again, by Definition 4.1,

$$\begin{aligned} L(\tilde{c}|_{[a,b]}) &= \int_a^b \|\tilde{c}'(u)\| du \\ &= \int_a^b \|c'(t(u))\| \left| \frac{dt}{du} \right| du \\ &= \int_{t(b)}^{t(a)} \|c'(t)\| dt \\ &= L(c|_{[t(b),t(a)]}). \end{aligned}$$

□

Definition 4.5 (Arclength parameter). A regular curve $c : I \rightarrow M$ is said to have an *arclength parameter* s if $\|c'(s)\| = 1$ for all $s \in I$. ■

If $c : I \rightarrow M$ has an arclength parameter, then

$$L(c|_{[a,b]}) = \int_a^b \|c'(s)\| ds = \int_a^b 1 ds = b - a, \quad \forall a, b \in I (a < b).$$

Meaning the "elapsed time" $b - a$ equals the distance traveled along $L(c|_{[a,b]})$.

For any regular curve $c : I \rightarrow M$ and $t_0 \in I$, let

$$s(t) := \int_{t_0}^t \|c'(t)\| dt.$$

Since $\frac{ds}{dt} = \|c'(t)\| > 0 \implies$ there exists a global inverse $t = t(s)$ by Theorem 4.3 (inverse function theorem).

Lemma 4.6. $\tilde{c}(s) = c(t(s))$ has an arclength parameter s .

Proof. By $\tilde{c}'(s) = c'(t(s))\frac{dt}{ds}$, we have

$$\begin{aligned} \|\tilde{c}'(s)\| &= \|c'(t(s))\| \left| \frac{dt}{ds} \right|, \\ \left| \frac{dt}{ds} \right| &= \frac{1}{\frac{ds}{dt}} = \frac{1}{\|c'(t)\|}. \end{aligned}$$

Meaning we have $\|\tilde{c}'(s)\| = 1$ for all $s \in I$. $\square$

Definition 4.7 (Piecewise smooth curve). A continuous curve $c : [a, b] \rightarrow M$ is a *piecewise smooth regular curve* if there exists a finite sequence $a = t_0 < t_1 < \dots < t_k = b$ such that

$$c|_{[t_i, t_{i+1}]} : [t_i, t_{i+1}] \rightarrow M$$

is regular for all $i = 0, \dots, k-1$.

Furthermore, the length $L(c)$ is defined by

$$L(c) := \sum_{i=0}^{k-1} L(c|_{[t_i, t_{i+1}]}) = \sum_{i=0}^{k-1} \int_{t_i}^{t_{i+1}} \|c'(t)\| dt.$$

■

Assume M is connected, i.e. any two points are joinable by a regular curve.

Definition 4.8 (Riemannian distance). The *Riemannian distance* $d(p, q)$ between two points $p, q \in M$ is defined as

$$d(p, q) := \inf\{L(c) : c \text{ is a piecewise smooth regular curve from } p \text{ to } q\}.$$

$d : M \times M \rightarrow \mathbb{R}$ is called the *Riemannian distance function*. If there exists some function c attaining the "inf", then c is called a *minimal curve*. $\blacksquare$

Theorem 4.9. The function $d : M \times M \rightarrow \mathbb{R}$ satisfies the axioms of distance. That is, for all $p, q, r \in M$,

- (1) $d(p, q) = 0$ if and only if $p = q$,
- (2) $d(p, q) = d(q, p)$,
- (3) $d(p, r) \leq d(p, q) + d(q, r)$.

Lemma 4.10. $(U, \phi) : a \text{ chart}$, $D \subset \phi(U)$ a *cpt set*. $c : [a, b] \rightarrow \phi^{-1}(D) : a \text{ piecewise smooth regular curve}$, Then $\exists C = C((U, \phi), g_{ij}, D) > 0$ s.t for $\bar{c} := \phi \circ c : [a, b] \rightarrow \mathbb{R}^n$

$$C^{-1}L(\bar{c}) \leq L(c) \leq CL(\bar{c}), \text{ where}$$

$$L(\bar{c}) = \int_a^b \|\bar{c}'(t)\|_{\mathbb{R}^n} dt \text{ is the Euclidean length.}$$

Proof. Define a continuous function $f : \phi(U) \times \mathbb{R}^n \rightarrow \mathbb{R}$ by

$$f(x, v) := \sum_{i,j=1}^n v^i v^j g_{ij}(\phi^{-1}(x))$$

Since $D \times \mathbb{S}^{n-1} \subset \mathbb{R}^{2n}$ is cpt, $f|_{D \times \mathbb{S}^{n-1}}$ has max/min. Since $f > 0$, the min is positive.

$$\implies \exists C > 0 \text{ s.t. } c^{-2} \leq f \leq C^2 \text{ on } D \times \mathbb{S}^{n-1}$$

Thus, $\forall x \in D$ and $\forall v \in \mathbb{R}^n \setminus \{0\}$:

$$C^{-2} \leq f\left(x, \frac{v}{\|v\|_{\mathbb{R}^n}}\right) \leq C^2$$

Multiplying $\|v\|_{\mathbb{R}^n}^2$

$$C^{-2} \|v\|_{\mathbb{R}^n}^2 \leq \sum_{i,j=1}^n v^i v^j g_{ij}(\phi^{-1}(x)) \leq C^2 \|v\|_{\mathbb{R}^n}^2$$

Let $\bar{c} = (c^1(t), \dots, c^n(t)) \implies c'(t) = \sum_{i=1}^n \frac{dc^i}{dt} \partial_i$.

$$\bar{c}'(t) = \left(\frac{dc^1}{dt}, \dots, \frac{dc^n}{dt} \right). \text{ Set } v := \bar{c}'(t)$$

$$C^{-2} \|\bar{c}'(t)\|_{\mathbb{R}^n}^2 \leq \|c'(t)\|^2 \leq C^2 \|\bar{c}'(t)\|_{\mathbb{R}^n}^2$$

Taking square roots and integrating with respect to t yields the result. $\square$

Lemma 4.11. *Let (U, ϕ) be a chart of M . Choose $x_0 \in \phi(U)$ and $r > 0$ s.t. $D := \bar{B}(x_0, 3r) \subset \phi(U)$. Let $C > 0$ be the constant from Lemma 4.10. Then:*

- (i) $\forall p, q \in \phi^{-1}(B(x_0, r))$:
 $C^{-1} \|\phi(p) - \phi(q)\|_{\mathbb{R}^n} \leq d(p, q) \leq C \|\phi(p) - \phi(q)\|_{\mathbb{R}^n}.$
- (ii) $\forall p \in \phi^{-1}(B(x_0, r)), \forall q \in M \setminus \phi^{-1}(D) : d(p, q) \leq 2C^{-1}r$

Proof. (1) $p, q \in \phi^{-1}(B(x_0, r))$. $\bar{\gamma}$: the line segment in $\mathbb{R}^n$ connecting $\phi(p)$ and $\phi(q)$. Since $B(x_0, r)$ is convex, $\text{gamma} \subset B(x_0, r)$. Set $\gamma := \phi^{-1} \circ \bar{\gamma}$. By Lemma 4.10,

$$\|\phi(p) - \phi(q)\|_{\mathbb{R}^n} \leq L(\bar{\gamma}) \leq C^{-1} L(\gamma) \geq C^{-1} d(p, q).$$

Next, take any piecewise smooth regular curve $c : [0, l] \rightarrow M$ from p to q . If $c([0, l]) \subset \phi^{-1}(D)$, let $\bar{c} := \phi \circ c$. By Lemma 4.10,

$$L(c) \geq C^{-1} L(\bar{c}) \geq C^{-1} \|\phi(p) - \phi(q)\|_{\mathbb{R}^n}.$$

If $c([0, l]) \not\subset \phi^{-1}(D)$, let

$$t_1 := \sup\{t \in [0, l] | c([0, t]) \subset \phi^{-1}(D)\}$$

and $c_1 := c|_{[0, t_1]}$. By Lemma 1.14, $c_1(t_1) \in \phi^{-1}(\partial D)$. Let $\bar{c}_1 := \phi \circ c_1$. By Lemma 4.10,

$$L(c) \geq L(c_1) \geq C^{-1}L(\bar{c}_1).$$

Since $\phi(p), \phi(q) \in B(x_0, r)$ and $\bar{c}_1(t_1) \in \partial B(x_0, 3r)$

$$\begin{aligned} L(\bar{c}_1) &\geq 2r > \|\phi(p) - \phi(q)\|_{\mathbb{R}^n} \\ \implies L(c) &> C^{-1}\|\phi(p) - \phi(q)\|_{\mathbb{R}^n} \text{ in both cases} \end{aligned}$$

Taking inf over c (using the fact that C in Lemma 4.10 is independent of c)

$$d(p, q) \geq C^{-1}\|\phi(p) - \phi(q)\|_{\mathbb{R}^n}$$

(2) Prf is similar to (1) above (omitted). $\square$

Proof of 4.9. (1) $p = q \implies$ taking $c(t) = p$ gives $L(c) = 0$ thus $d(p, q) = 0$. conversely, assume $d(p, q) = 0$. Fix a chart (U, ϕ) around p and let $x_0 := \phi(p)$. Choose $r > 0$ s.t $D_r := \bar{B}(x_0, 3r) \subset \phi(U)$. By 4.11,

$$\exists C_r > 0 \text{ s.t if } q \in M \setminus \phi^{-1}(D_r), \text{ then } d(p, q) \geq 2C_r^{-1}r$$

Contradicting $d(p, q) = 0$. Thus $q \in \phi^{-1}(D_r)$. Since $r > 0$ is arbitrary, $p = q$

(2) Let c be a piecewise smooth regular curve from p to q . Reversing, the parameter yields a curve from q to p with the same length. $\implies d(p, q) = d(q, p)$

(3) Let c_{pq} be a curve from p to q , and c_{qr} from q to r . Concatenating them yields a piecewise smooth regular curve from p to r . $d(p, r) \leq L(c_{pqr}) = L(c_{pq}) + L(c_{qr})$ Taking inf over c_{pq} and $c_{qr} \implies$ triangle inequality. $\square$

Remark 4.12. Hausdorff property is essential for $d(p, q) = 0$. In non-Hausdorff manifolds, $d(p, q) = 0$ does not necessarily imply $p = q$ $\blacksquare$

Problem 7. Define $L, R_+, R_- \subset \mathbb{R}^2$ as:

$$\begin{aligned} L &:= \{(x, 0) \in \mathbb{R}^2 | x < 0\} \\ R_+ &:= \{(x, 1) \in \mathbb{R}^2 | x \geq 0\}, R_- := \{(x, -1) \in \mathbb{R}^2 | x \geq 0\} \end{aligned}$$

$M := L \cup R_+ \cup R_-$, $U := L \cup R_+$, $V := L \cup R_-$. Let $\pi_1 : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the projection to the 1st component and $\phi := \pi_1|_U : U \rightarrow \mathbb{R}$, $\psi := \pi_1|_V : V \rightarrow \mathbb{R}$. $W \subset M$ is open $\iff \phi(W \cap U), \psi(W \cap V) \subset \mathbb{R}$ are open. This makes M a topological space s.t. ϕ, ψ are homeomorphisms,

$$\phi(U \cap V) = \psi(U \cap V) = (-\infty, 0), \text{ and } \psi \circ \phi^{-1} = Id_{(-\infty, 0)}$$

- (i) Show M is not Hausdorff
- (ii) Show $d((0, 1), (0, -1)) = 0$ for any riemann metric on M $\blacksquare$

Definition 4.13. For $p \in M$, $r > 0$

$$B(p, r) := \{q \in M | d(p, q) < r\}$$

is called a geodesic ball of radius r centered at p $\blacksquare$

Proposition 4.14. *For a subset $O \subset M$ the following are equivalent:*

- (i) O is open in M
- (ii) $\forall p \in O, \exists \delta > 0$ s.t. $B(p, \delta) \subset O$

Proof. Take a chart (U, ϕ) around p and let $x_0 := \phi(p)$. Since $\phi(O \cap U) \subset \mathbb{R}^n$ is open, $\exists \gamma > 0$ s.t. $D_i = \bar{B}(x_0, 3r) \subset \phi(O \cap U)$. For the constant $C > 0$ from 4.11, choose $\delta > 0$ s.t. $0 < \delta < 2C^{-1}r$. Then $\forall q \in M \setminus \phi^{-1}(D)$, $d(p, q) \geq 2C^{-1}r > \delta$. Thus $q \in B(p, \delta) \implies q \in \phi^{-1}(D)$, we have $B(p, \delta) \subset O$. (2) $\implies$ (1) : Let $(U_\alpha, \phi_\alpha)_{\alpha \in \Lambda}$ be an atlas of M . Then $O = \bigcup_{\alpha \in \Lambda} (O \cap U_\alpha)$. Since each $\phi_\alpha : U_\alpha \rightarrow \phi_\alpha(U_\alpha)$ is a homeo, $O \cap U_\alpha$ is open in $U_\alpha \equiv \phi_\alpha(O \cap U_\alpha)$ is open in $\phi_\alpha(U_\alpha)$. Thus it suffices to show: $\phi(O \cap U) \subset \mathbb{R}^n$ is open for any chart (U, ϕ)

$x \in \phi(O \cap U)$, set $p := \phi^{-1}(x) \in O \cap U$. By assumption, $\exists r_1 > 0$ s.t. $B(p, r_1) \subset O$. Since $\phi(U) \subset \mathbb{R}^n$ is open, $\exists r_2 > 0$ s.t. $\bar{B}(x, 3r_2) \subset \phi(U)$. For (U, ϕ) and $D := \bar{B}(x, 3r_2)$ let $C > 0$ be the constant from Lemma 4.10. Set $r := \min C^{-1}r_1, r_2$. Then $B(p, Cr) \subset B(p, r_1) \subset O$. For any $y \in B(x, r)$ set $q := \phi^{-1}(y) \in \phi^{-1}(B(x, r_2)) \subset U$. By Lemma 4.10:

$$d(p, q) \leq C\|x - y\|_{\mathbb{R}^n} < Cr,$$

Thus $q \in B(p, Cr) \subset O$, which implies $q \in O \cap U$ and $y \in \phi(O \cap U)$. Therefore $B(x, r) \subset \phi(O \cap U)$ and $\phi(O \cap U)$ is open in $\mathbb{R}^n$ $\square$

Problem 8. Using 4.14, show that for any $p \in M$ and $r > 0$, the geodesic ball $B(p, r)$ is an open set in M . The following properties hold for distance functions on general metric spaces, thus proofs are omitted. $d : M \times M \rightarrow \mathbb{R}$.

Proposition 4.15. $\forall p_0 \in M, f(p) := d(p, p_0)$ is continuous

Proposition 4.16. *For a sequence $p_{i=1}^\infty \subset M$ and $p \in M$, the following are equivalent:*

- (i) $p_i \rightarrow p$ as $i \rightarrow \infty$
- (ii) $d(p_i, p) \rightarrow 0$ as $i \rightarrow \infty$

Basic topological properties of geodesic balls (proofs omitted:)

Proposition 4.17. $\forall p \in M$ and $r > 0$

- (i) $\partial B(p, r) = \{q \in M \mid d(p, q) = r\}$
- (ii) $\bar{B}(p, r) := \overline{B(p, r)} = \{q \in M \mid d(p, q) \leq r\}$

■

5 Levi-Civita connection

Assume M is a Riemannian manifold.

Definition 5.1 (Levi-Civita connection).

$$\nabla : \underset{\in (X, Y)}{\mathfrak{X}(\mathcal{M})} \times \underset{\in \nabla \times Y}{\mathfrak{X}(M)} \rightarrow \underset{\in \nabla \times Y}{(M)}$$

is called the *covariant derivative* or *Levi-Civita connection* if $\forall X, Y, Z \in \mathfrak{X}(M)$ and $\forall f \in C^\infty(M)$:

- (1) $\nabla_{X+Y}Z = \nabla_XZ + \nabla_YZ$
- (2) $\nabla_{fX}Y = f\nabla_XY$
- (3) $\nabla_X(Y+Z) = \nabla_XY + \nabla_XZ$
- (4) $\nabla_X(fY) = X(f)Y + f\nabla_XY$
- (5) $\nabla_XY - \nabla_YX = [X, Y]$ (torsion-free)
- (6) $X\langle Y, Z \rangle = \langle \nabla_XY, Z \rangle + \langle Y, \nabla_XZ \rangle$ (metric compatibility)

■

Theorem 5.2 (Fundamental theorem of Riemannian geometry). *There exists a unique (!) Levi-Civita connection ∇ on a Riemannian manifold M .*

Proof. We start by showing uniqueness. By (6):

$$\begin{aligned} X\langle Y, Z \rangle &= \langle \nabla_XY, Z \rangle + \langle Y, \nabla_XZ \rangle \\ Y\langle Z, X \rangle &= \langle \nabla_YZ, X \rangle + \langle Z, \nabla_YX \rangle \\ -Z\langle X, Y \rangle &= -\langle \nabla_ZX, Y \rangle - \langle X, \nabla_ZY \rangle \end{aligned}$$

Summing over these and applying (5):

$$\begin{aligned} \langle \nabla_XY, Z \rangle &= \frac{1}{2} \{ X\langle Y, Z \rangle + Y\langle Z, X \rangle - Z\langle X, Y \rangle + \langle [X, Y], Z \rangle \\ &\quad - \langle [Y, Z], X \rangle + \langle [Z, X], Y \rangle \} \end{aligned} \quad (5.1)$$

This shows uniqueness.

We now show existence. Let the right-hand side of (5.1) be $F_{X,Y}(Z)$. Then $\forall f_1, f_2 \in C^\infty(M)$ and $\forall Z_1, Z_2 \in \mathfrak{X}(M)$:

$$F_{X,Y}(f_1Z_1 + f_2Z_2) = f_1F_{X,Y}(Z_1) + f_2F_{X,Y}(Z_2),$$

is easily verified. For a chart (U, ϕ) , we can define

$$\nabla_X^{(U, \phi)}Y := \sum_{i,j=1}^n F_{X,Y}(\partial_i)g^{ij}\partial_j,$$

where $(g^{ij}) = (g_{ij})^{-1}$. For $Z = \sum_{k=1}^n Z^k\partial_k$, using that $\sum_{j=1}^n g^{ij}g_{jk} = \delta_k^i$, we have

$$\begin{aligned} \langle \nabla_X^{(U, \phi)}Y, Z \rangle &= \sum_{i,j,k=1}^n F_{X,Y}(\partial_i)g^{ij}Z^k g_{jk} \\ &= \sum_{i=1}^n F_{X,Y}(\partial_i)Z^i = F_{X,Y}(Z). \end{aligned}$$

Note that this is independent of the choice of chart (U, ϕ) ! This implies that for any other chart (V, ψ) such that $U \cap V \neq \emptyset$, $\nabla_X^{U, \phi} = \nabla_X^{(V, \psi)}$. Thus $\nabla_XY := \nabla_X^{(U, \phi)}Y$ yields a globally defined vector field. Direct calculation verifies all axioms (1)–(6) of Definition 5.1. $\square$

Problem 9. Verify axioms (1)–(6) of Definition 5.1 for the Levi-Civita connection ∇ constructed in Theorem 5. ■

Definition 5.3 (Christoffel symbols). For a chart (U, ϕ) , we define functions Γ_{ij}^k by

$$\nabla_{\partial_i} \partial_j = \sum_{k=1}^n \Gamma_{ij}^k \partial_k$$

called *Christoffel symbols*. ■

Remark 5.4. ∇ is NOT a tensor field, implying that Γ_{ij}^k are not the components of a tensor. ■

Proposition 5.5. $\forall i, j, k = 1, \dots, n$:

- (1) $\Gamma_{ij}^k = \sum_{l=1}^n \langle \nabla_{\partial_i} \partial_j, \partial_l \rangle g^{lk} = \frac{1}{2} \sum_{l=1}^n g^{kl} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$.
- (2) $\Gamma_{ij}^k = \Gamma_{ji}^k$.
- (3) $\partial_k g_{ij} = \frac{\partial g_{ij}}{\partial x^k} = \sum_{l=1}^n (\Gamma_{ik}^l g_{lj} + \Gamma_{jk}^l g_{li})$.

Proof. By Definition 5.3: $\langle \nabla_{\partial_i} \partial_j, \partial_l \rangle = \Gamma_{ij}^k g_{kl}$. Multiplying by g^{lm} and summing over l yields the first equality (1). The second equality follows from Definition 5.1 with $X = \partial_i$, $Y = \partial_j$, $Z = \partial_l$ and $[\partial_i, \partial_j]$.

(2) follows from (1) and the symmetry of g_{ij} .

(3) follows from Definition 5.1 with $X = \partial_k$, $Y = \partial_i$, $Z = \partial_j$ and (2). □

For local expressions $X = \sum_{i=1}^n X^i \partial_i$ and $Y = \sum_{j=1}^n Y^j \partial_j$, calculation using Definition 5.1 (1)–(4) yields

$$\begin{aligned} \nabla_X Y &= \sum_{i=1}^n X^i \nabla_{\partial_i} \left(\sum_{j=1}^n Y^j \partial_j \right) \\ &= \sum_{i=1}^n X^i \left(\sum_{j=1}^n (\partial_i Y^j \partial_j + Y^j \sum_{k=1}^n \Gamma_{ij}^k \partial_k) \right) \\ &= \sum_{k=1}^n \left\{ X(Y^k) + \sum_{i,j=1}^n \Gamma_{ij}^k X^i Y^j \right\} \partial_k. \end{aligned}$$

A key fact is that $(\nabla_X Y)_p$ depends only on X_p and the behaviour of Y along X_p direction at p . Thus, for $v \in T_p M$, we define

$$\nabla_v Y := (\nabla_X Y)_p \text{ such that } X_p = v,$$

which is well-defined and independent of the extension X .

Lemma 5.6. Let (U, ϕ) and (V, ψ) be charts such that $U \cap V \neq \emptyset$. Let $\phi = (x^1, \dots, x^n)$ and $\psi = (y^1, \dots, y^n)$ and $\Gamma_{ij}^k, \hat{\Gamma}_{ij}^k$ be Christoffel symbols with respect to each chart. Then $\forall \alpha, \beta, \gamma = 1, \dots, n$:

$$\hat{\Gamma}_{\alpha\beta}^\gamma = \sum_{k=1}^n \left(\frac{\partial^2 x^k}{\partial y^\alpha \partial y^\beta} + \sum_{i,j=1}^n \frac{\partial x^i}{\partial y^\alpha} \frac{\partial x^j}{\partial y^\beta} \Gamma_{ij}^k \right) \frac{\partial y^\gamma}{\partial x^k}.$$

Proof. By the transformation law $\frac{\partial}{\partial y^\alpha} = \sum_{i=1}^n \frac{\partial x^i}{\partial y^\alpha} \frac{\partial}{\partial x^i}$:

$$\nabla_{\frac{\partial}{\partial y^\alpha}} \frac{\partial}{\partial y^\beta} = \sum_{i,j=1}^n \left(\frac{\partial x^i}{\partial y^\alpha} \left(\frac{\partial}{\partial x^i} \frac{\partial x^j}{\partial y^\beta} \right) \frac{\partial}{\partial x^j} + \frac{\partial x^i}{\partial y^\alpha} \frac{\partial x^j}{\partial y^\beta} \sum_{k=1}^n \Gamma_{ij}^k \frac{\partial}{\partial x^k} \right).$$

Using $\frac{\partial}{\partial x^i} = \sum_{\gamma=1}^n \frac{\partial y^\gamma}{\partial x^i} \frac{\partial}{\partial y^\gamma}$:

$$\sum_{i=1}^n \frac{\partial x^i}{\partial y^\alpha} \left(\frac{\partial}{\partial x^i} \frac{\partial x^j}{\partial y^\beta} \right) = \sum_{i,\gamma=1}^n \frac{\partial x^i}{\partial y^\alpha} \frac{\partial y^\gamma}{\partial x^i} \frac{\partial^2 x^j}{\partial y^\gamma \partial y^\beta} = \frac{\partial^2 x^j}{\partial y^\alpha \partial y^\beta}.$$

Note that $\frac{\partial}{\partial x^i} \frac{\partial x^j}{\partial y^\beta} = \frac{\partial}{\partial y^\beta} \left(\frac{\partial x^j}{\partial x^i} \right) = \frac{\partial}{\partial y^\beta} \delta_i^j = 0$. Thus,

$$\begin{aligned} \sum_{\gamma=1}^n \hat{\Gamma}_{\alpha\beta}^\gamma \frac{\partial}{\partial y^\gamma} &= \nabla_{\frac{\partial}{\partial y^\alpha}} \frac{\partial}{\partial y^\beta} \\ &= \sum_{k=1}^n \left(\frac{\partial^2 x^k}{\partial y^\alpha \partial y^\beta} + \sum_{i,j=1}^n \frac{\partial x^i}{\partial y^\alpha} \frac{\partial x^j}{\partial y^\beta} \Gamma_{ij}^k \right) \frac{\partial}{\partial x^k} \\ &= \sum_{\gamma=1}^n \sum_{k=1}^n \left(\frac{\partial^2 x^k}{\partial y^\alpha \partial y^\beta} + \sum_{i,j=1}^n \frac{\partial x^i}{\partial y^\alpha} \frac{\partial x^j}{\partial y^\beta} \Gamma_{ij}^k \right) \frac{\partial y^\gamma}{\partial x^k} \frac{\partial}{\partial y^\gamma}. \end{aligned}$$

□

6 Covariant derivative along curves and maps

6.1 Covariant derivative along curves

Let M be a Riemannian manifold and $c : I \rightarrow M$ a C^∞ curve (not necessarily regular).

Definition 6.1 (C^∞ vector field along a curve). Let $X : \begin{matrix} t \mapsto \\ \in I \end{matrix} \begin{matrix} X(t) \\ \in T_{c(t)} M \end{matrix}$ be a vector field along c . Then X is C^∞ if $\forall$ charts (U, ϕ) , the components $X^i(t)$ in

$$X(t) = \sum_{i=1}^n X^i(t) (\partial_i)_{c(t)}, t \in c^{-1}(U),$$

are C^∞ . ■

Henceforth, vector fields along curves are assumed to be C^∞ , unless otherwise stated.

Definition 6.2. The *covariant derivative* of a vector field X along a curve c is defined on a chart (U, ϕ) by

$$(\nabla_c X)(t) := \sum_{k=1}^n \left(\frac{dX^k}{dt} + \sum_{i,j=1}^n \Gamma_{ij}^k(c(t)) \frac{dc^i}{dt} X^j \right) (\partial_k)_{c(t)},$$

where $\phi(c(t)) = (c^1(t), \dots, c^n(t))$ and $X(t) = \sum_{i=1}^n X^i(t)(\partial_i)_{c(t)}$. ■

Remark 6.3. $\nabla_c X$ is a C^∞ vector field along c . ■

Lemma 6.4. $\nabla_c X$ is independent of the choice of chart (U, ϕ) .

Proof. Let (U, ϕ) and (V, ψ) be charts such that $c(t) \in U \cap V$. Let $\phi = (x^1, \dots, x^n)$, $\psi = (y^1, \dots, y^n)$, $\partial_i := \frac{\partial}{\partial x^i}$ and $\hat{\partial}_\alpha := \frac{\partial}{\partial y^\alpha}$. Then $\hat{\Gamma}_{\alpha\beta}^\gamma$ are Christoffel symbols with respect to (V, ψ) and Γ_{ij}^k are Christoffel symbols with respect to (U, ϕ) . Finally, let $\phi(c(t)) = (c^1(t), \dots, c^n(t))$ and $\psi(c(t)) = (\hat{c}^1(t), \dots, \hat{c}^n(t))$ and $X(t) = \sum_{i=1}^n X^i(t)(\partial_i)_{c(t)} = \sum_{\alpha=1}^n \hat{X}^\alpha(t)(\hat{\partial}_\alpha)_{c(t)}$.

It suffices to show that

$$\sum_{\gamma=1}^n \left(\frac{d\hat{X}^\gamma}{dt} + \sum_{\alpha, \beta=1}^n \hat{\Gamma}_{\alpha\beta}^\gamma \frac{d\hat{c}^\alpha}{dt} \hat{X}^\beta \right) \hat{\partial}_\gamma = \sum_{k=1}^n \left(\frac{dX^k}{dt} + \sum_{i, j=1}^n \Gamma_{ij}^k \frac{dc^i}{dt} X^j \right) \partial_k. \quad (6.1)$$

By the transformation laws: $\hat{\partial}_\gamma = \sum_{k=1}^n \frac{\partial x^k}{\partial y^\gamma} \partial_k$, $\hat{X}^\alpha = \sum_{i=1}^n \frac{\partial y^\alpha}{\partial x^i} X^i$ Differentiating the second with respect to t yields

$$\begin{aligned} \frac{d\hat{X}^\gamma}{dt} &= \sum_{i=1}^n \left[\frac{\partial y^\gamma}{\partial x^i} \frac{dX^i}{dt} + \left(\frac{d}{dt} \frac{\partial y^\gamma}{\partial x^i} \right) X^i \right] \\ &= \sum_{i=1}^n \left(\frac{dy^\gamma}{dx^i} \frac{dX^i}{dt} + X^i \sum_{j=1}^n \frac{\partial^2 y^\gamma}{\partial x^i \partial x^j} \frac{dc^j}{dt} \right). \end{aligned}$$

Similarly,

$$\frac{d\hat{c}^\alpha}{dt} = \sum_{i=1}^n \frac{\partial y^\alpha}{\partial x^i} \frac{dc^i}{dt}.$$

We also have

$$\sum_{\gamma=1}^n \frac{\partial^2 y^\gamma}{\partial x^i \partial x^j} \frac{\partial x^k}{\partial y^\gamma} + \sum_{\alpha, \beta=1}^n \frac{\partial^2 x^k}{\partial y^\alpha \partial y^\beta} \frac{\partial y^\alpha}{\partial x^i} \frac{\partial y^\beta}{\partial x^j} = 0. \quad (6.2)$$

Substituting into Lemma 5.6 and using (6.2) yields (6.1).

We now just have to prove (6.2).

$$\begin{aligned} 0 &= \frac{\partial}{\partial y^\beta} \delta_{ki} = \sum_{\alpha=1}^n \frac{\partial}{\partial y^\beta} \left(\frac{\partial x^k}{\partial y^\alpha} \frac{\partial y^\alpha}{\partial x^i} \right) \\ &= \sum_{\alpha=1}^n \left(\frac{\partial x^k}{\partial y^\alpha} \frac{\partial}{\partial y^\beta} \frac{\partial y^\alpha}{\partial x^i} + \frac{\partial^2 x^k}{\partial y^\alpha \partial y^\beta} \frac{\partial y^\alpha}{\partial x^i} \right) \\ &= \sum_{\gamma, l=1}^n \frac{\partial x^k}{\partial y^\gamma} \frac{\partial x^l}{\partial y^\beta} \frac{\partial^2 y^\gamma}{\partial x^i \partial x^l} + \sum_{\alpha=1}^n \frac{\partial^2 x^k}{\partial y^\alpha \partial y^\beta} \frac{\partial y^\alpha}{\partial x^i}. \end{aligned}$$

Multiplying by $\frac{\partial y^\beta}{\partial x^j}$ and summing over β yields (6.2). □

Proposition 6.5 (Properties of covariant derivative along curves). *Let X, Y be vector fields along $c : I \rightarrow M$, $f : I \rightarrow \mathbb{R}$ a C^∞ function and $a, b \in \mathbb{R}, t \in I$. Then*

- (1) $\nabla_c(aX + bY) = a\nabla_c X + b\nabla_c Y$.
- (2) $\nabla_c(fX) = \frac{df}{dt}X + f\nabla_c X$.
- (3) $\frac{d}{dt}\langle X, Y \rangle = \langle \nabla_c X, Y \rangle + \langle X, \nabla_c Y \rangle$.

Proof. (1) and (2) are trivial.

On a chart (U, ϕ) , with $\phi(c(t)) = (c^1(t), \dots, c^n(t))$, we have

$$X(t) = \sum_{i=1}^n X^i(t)(\partial_i)_{c(t)}, Y(t) = \sum_{j=1}^n Y^j(t)(\partial_j)_{c(t)}.$$

Then (by the chain and product rules),

$$\begin{aligned} \frac{d}{dt}\langle X, Y \rangle &= \sum_{i,j=1}^n \frac{d}{dt} (g_{ij}(c(t))X^i(t)Y^j(t)) \\ &= \sum_{i,j=1}^n \left(\frac{dX^i}{dt}Y^j g_{ij} + X^i \frac{dY^j}{dt} g_{ij} + X^i Y^j \sum_{k=1}^n \frac{\partial g_{ij}}{\partial x^k} \frac{dc^k}{dt} \right). \end{aligned}$$

On the other hand,

$$\begin{aligned} \langle \nabla_c X, Y \rangle &= \left\langle \sum_{i=1}^n \left(\frac{dX^i}{dt} + \sum_{k,l=1}^n \Gamma_{kl}^i \frac{dc^k}{dt} X^l \right) \partial_i, \sum_{l=1}^n Y^l \partial_l \right\rangle \\ &= \sum_{i,j=1}^n \left(\frac{dX^i}{dt} Y^j g_{ij} + \sum_{k,l=1}^n \Gamma_{kl}^i \frac{dc^k}{dt} X^l Y^j g_{ij} \right). \\ \langle X, \nabla_c Y \rangle &= \sum_{i,j=1}^n \left(X^i \frac{dY^j}{dt} g_{ij} + \sum_{k,l=1}^n \Gamma_{kl}^j \frac{dc^k}{dt} X^i Y^l g_{ij} \right). \end{aligned}$$

Combining these with Proposition 5.5 (3) yields (3). $\square$

Definition 6.6. A vector field X along $C : I \rightarrow M$ is parallel if $\nabla_C X = 0$ ($\forall t \in I$). $\blacksquare$

Lemma 6.7. X, Y are parallel vector fields along $C \implies \langle X(t), Y(t) \rangle$ is independent of $t \in I$.

Proof. (3):

$$\frac{d}{dt}\langle X(t), Y(t) \rangle = \langle \nabla_C X(t), Y(t) \rangle + \langle X(t), \nabla_C Y(t) \rangle$$

$\square$

Proposition 6.8. $\forall t_0 \in I$ and $\forall v \in T_{C(t_0)}M$, $\exists!$ parallel vector field X along C such that $X(t_0) = v$.

Proof. On a chart (U, ϕ) , let

$$\phi(c(t)) = (c^1(t), \dots, c^n(t)), X(t) = \sum_{i=1}^n X^i(t)(\partial_i)_{c(t)}$$

Local expression of $\nabla_C X = 0$:

$$\frac{dX^k}{dt} + \sum_{i,j=1}^n \Gamma_{ij}^k \frac{dc^i}{dt} X^j = 0, k = 1, \dots, n$$

this is a First-order linear ODE system. By existence and uniqueness of ODEs, given $v = \sum X^k(t_0)(\partial_k)$, a unique solution exists locally. Since the equation $\nabla_C X = 0$ is independent of coordinates, local solutions patch together to yield $X(t) \forall t \in I$. $\square$

Definition 6.9. $\forall t_0, t \in I$ and $\forall v \in T_{c(t_0)}M$, Let X be the parallel vector field along c such that $X(t_0) = v$. Define $P_{c(t)}v := X(t)$ the map

$$P_{c(t)} : T_{c(t_0)}M \rightarrow T_{c(t)}M$$

Is the parallel transport along c . $\blacksquare$

Proposition 6.10. $\forall t_0, t \in I$, the parallel transport $P_{c(t)} : T_{c(t_0)}M \rightarrow T_{c(t)}M$ is a linear isometry.

Proof. Let $a, b \in \mathbb{R}$ and $v, w \in T_{c(t_0)}M$. Let X, Y be parallel vector fields along c such that $X(t_0) = v$ and $Y(t_0) = w$.

$$\implies X(t) = P_{c(t)}v, Y(t) = P_{c(t)}w.$$

By (1) $\nabla_c(aX + bY) = a\nabla_c X + b\nabla_c Y = 0$. $aX + bY$ is parallel along c with $(aX + bY)(t_0) = av + bw$. Thus, $P_{c(t)}(av + bw) = (aX + bY)(t) = aP_{c(t)}v + bP_{c(t)}w$. $P_{c(t)}$ is linear. Isometry follows from Lemma 6.7. $\langle P_{c(t)}v, P_{c(t)}w \rangle = \langle X(t), Y(t) \rangle = \langle X(t_0), Y(t_0) \rangle = \langle v, w \rangle$. $\square$

Proposition 6.11. For any $t \in I$ and a C^∞ vector field Y along c :

$$(\nabla_c Y)(t) = \lim_{h \rightarrow 0} \frac{P_{c(t)}Y(t+h) - Y(t)}{h}$$

where the limit is with respect to the standard top. on $T_{c(t)}M$.

Proof. Fix $t \in I$ and let $e_{i=1}^\infty$ be a basis of $T_{c(t)}M$. Let $E_i(t) = e_i \implies P_{c(t')}e_i = E_i(t') \forall t' \in I$. By Proposition 6.10, $\{E_i(t')\}_{i=1}^n$ is a basis of $T_{c(t')}M$. Express Y as $Y(t') = \sum_{i=1}^n Y^i(t')E_i(t')$. Since $\nabla_c E_i = 0$,

$$\nabla_c Y = \sum_{i=1}^n \left(\frac{dY^i}{dt} E_i + Y^i \nabla_c E_i \right) = \sum_{i=1}^n \frac{dY^i}{dt} E_i$$

Conversely, by uniqueness of parallel transport $P_{c(t)}E_i(t+h) = E_i(t)$:

$$\begin{aligned} P_{c(t)}Y(t+h) &= \sum_{i=1}^n Y^i(t+h)P_{c(t)}E_i(t+h) \\ &= \sum_{i=1}^n Y^i(t+h)E_i(t). \end{aligned}$$

Thus

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{P_{c(t)}Y(t+h) - Y(t)}{h} &= \lim_{h \rightarrow 0} \sum_{i=1}^n \frac{Y^i(t+h) - Y^i(t)}{h} E_i(t) \\ &= \sum_{i=1}^n \frac{dY^i}{dt}(t) E_i(t) \end{aligned}$$

□

6.2 Covariant derivative along maps

Let M be an n -dimensional Riemannian manifold, N a differentiable manifold, $\xi : N \rightarrow M$ a C^∞ map.

Definition 6.12. A map $X : N \ni p \rightarrow X_p \in T_{\xi(p)}M$ is a vector field along ξ , X is C^∞ if $\forall$ chart (U, ϕ) of M and $\forall (V, \psi)$ of N such that $\xi(V) \subset U$, the components $X^i : V \rightarrow \mathbb{R}$ in $X_p = \sum_{i=1}^n X^i(p)(\partial_i)_{\xi(p)}$, $p \in V$ are C^∞ , where $\phi = (x^1, \dots, x^n)$ and $\partial_i := \frac{\partial}{\partial x^i}$. Henceforth, vector fields along maps are assumed C^∞ unless otherwise noted. ■

Definition 6.13. (1) For a vector field X on M ,

$$X \circ \xi : N \ni p \rightarrow X_{\xi(p)} \in T_{\xi(p)}M$$

defines a vector field $X \circ \xi$ along ξ .

(2) For a vector field Y on N $d\xi(Y) : N \ni p \rightarrow d\xi_p(Y_p) \in T_{\xi(p)}M$, defines a vector field $d\xi(Y)$ along ξ . ■

Definition 6.14. Let X be a vector field along a C^∞ map $\xi : N \rightarrow M$ and $v \in T_p N$ ($p \in N$). Let $c : I \rightarrow N$ be a C^∞ curve such that $c(0) = p$, $c'(0) = v$. Define the covariant derivative $\nabla_v^\xi X$ of X in the direction v as

$$\nabla_v^\xi := (\nabla_{\xi \circ c} X \circ c)(0) \in T_{\xi(p)}M$$

where the right hand side is the covariant derivative of the vector field $X \circ c$ along the curve $\xi \circ c : I \rightarrow M$ ■

Lemma 6.15. Let X be a vector field along ξ locally expressed as $X_p = \sum_{i=1}^n X^i(p)(\partial_i)_{\xi(p)}$ on any chart of M . Then $\nabla_v^\xi X = \sum_{i=1}^n \{(vX^i)(\partial_i)_{\xi(p)} + X^i(p)\nabla_{d\xi_p(v)}\partial_i\}$ In particular, $\nabla_v^\xi X$ is well-defined (independent of c)

Proof. Using parallel transport

$$\begin{aligned}
P_{\xi \circ c(0)} : T_{\xi \circ c(t)} M &\rightarrow T_{\xi(p)} M \text{ along } \xi \circ c \\
P_{\xi \circ c(0)} X_{c(t)} - X_p &= \sum_{i=1}^n \{X^i(c(t)) P_{\xi \circ c(0)} (\partial_i)_{\xi \circ c(t)} - X^i(p) (\partial_i)_{\xi(p)}\} \\
&= \sum_{i=1}^n [\{X^i(c(t)) - X^i(p)\} P_{\xi \circ c(0)} (\partial_i)_{\xi \circ c(t)} + X^i(p) \{P_{\xi \circ c(0)} (\partial_i)_{\xi \circ c(t)} - (\partial_i)_{\xi(p)}\}]
\end{aligned}$$

Dividing by t , taking $t \rightarrow 0$, and applying Proposition 6.10

$$\nabla_v^\xi X = (\nabla_{\xi \circ c} X \circ c)(0) = \sum_{i=1}^n \{(vX^i)(\partial_i)_{\xi(p)} + X^i(p) \lim_{t \rightarrow 0} \frac{P_{\xi \circ c(0)} (\partial_i)_{\xi \circ c(t)} - (\partial_i)_{\xi(p)}}{t}\}$$

Since $(\xi \circ c)'(0) = d\xi_p(c'(0)) = d\xi_p(v)$, we have $(\nabla_{\xi \circ c} \partial_i)(0) = \nabla_{d\xi_p(v)} \partial_i$. $\square$

Proposition 6.16. *Let $v, w \in T_p N$ ($p \in N$), X, Y be vector fields along ξ , $f : N \rightarrow \mathbb{R}$ be a C^∞ function. and $a, b \in \mathbb{R}$. Then*

- (i) $\nabla_{av+bw}^\xi X = a\nabla_v^\xi X + b\nabla_w^\xi X$.
- (ii) $\nabla_v^\xi (aX + bY) = a\nabla_v^\xi X + b\nabla_v^\xi Y$.
- (iii) $\nabla_v^\xi (fX) = (vf)X_p + f(p)\nabla_v^\xi X$.
- (iv) For a vector field Z on M , $\nabla_v^\xi (Z \circ \xi) = \nabla_{d\xi_p(v)} Z$.

Proposition 6.17. (1) For a vector field X, Y on N ,

$$\nabla_x^\xi (d\xi(Y)) - \nabla_Y^\xi (d\xi(X)) = d\xi([X, Y])$$

(2) For $v \in T_p N$ ($p \in N$) and vector field X, Y along ξ $v\langle X, Y \rangle = \langle \nabla_v^\xi X, Y_p \rangle + \langle X_p, \nabla_v^\xi Y \rangle$.

Proof. (1) On a chart around $\xi(p)$, locally exepress

$$\begin{aligned}
d\xi(X)_p &= \sum_{i=1}^n X^i(p) (\partial_i)_{\xi(p)} \\
d\xi(Y)_p &= \sum_{i=1}^n Y^i(p) (\partial_i)_{\xi(p)}
\end{aligned}$$

By Lemma 6.15 and $\Gamma_{ij}^k = \Gamma_{ji}^k$,

$$\begin{aligned}
\nabla_X^\xi &= \sum_{k=1}^i \{XY^k + \sum_{i,j=1}^n \Gamma_{ij}^k X^i x^j\} \partial_k \\
\nabla_Y^\xi &= \sum_{k=1}^i \{YX^k + \sum_{i,j=1}^n \Gamma_{ij}^k Y^i x^j\} \partial_k
\end{aligned}$$

$\nabla_X^\xi(d\xi(Y)) - \nabla_Y^\xi(d\xi(X)) = \sum_{k=1}^n (XY^k - YX^k)\partial_k$. Let x^k be coordinate components

$$\begin{aligned} d\xi([X, Y])(x^k) &= [X, Y](x^k \circ \xi) \\ &= X(Y(x^k \circ \xi)) - Y(X(x^k \circ \xi)) \\ &= X(d\xi(Y)x^k) - Y(d\xi(X)x^k) \\ &= XY^k - YX^k \end{aligned}$$

Hence $d\xi([X, Y]) = \sum_{k=1}^n d\xi([X, Y])(x^k)\partial_k = \sum_{k=1}^n (XY^k - YX^k)\partial_k$.

(2) Let $C : I \rightarrow N$ be a C^∞ curve such that $C(0) = p, C'(0) = v$. Set $X(t) = X_{C(t)}, Y(t) = Y_{C(t)}$. (Vector field along $\xi \circ C$). Parallel transports

$$\begin{aligned} P_{\xi \circ C(t)} : T_{\xi \circ C(0)}M &\rightarrow T_{\xi \circ C(t)}M \\ P_{\xi \circ C(0)} : T_{\xi \circ C(t)}M &\rightarrow T_{\xi \circ C(0)}M \end{aligned}$$

are inverse isometries to each other. (cf. Proposition 6.10) Thus

$$\begin{aligned} &\langle X(t), Y(t) \rangle - \langle X(0), Y(0) \rangle \\ &= \langle X(t), Y(t) \rangle - \langle X(0), Y(0) \rangle - \langle P_{\xi \circ C(0)}X(t), Y(t) \rangle + \langle X(0), P_{\xi \circ C(0)}Y(t) \rangle \\ &= \langle X(t) - P_{\xi \circ C(0)}X(t), Y(t) \rangle + \langle X(0), P_{\xi \circ C(0)}Y(t) - Y(0) \rangle \\ &= \langle P_{\xi \circ C(0)}X(t) - X(0), P_{\xi \circ C(0)}Y(t) \rangle + \langle X(0), P_{\xi \circ C(0)}Y(t) - Y(0) \rangle \end{aligned}$$

Divide by t , let $t \rightarrow 0$, and apply Proposition 6.10 □

7 First Variation Formular and Geodesics

Consider a variation of a C^∞ curve $c : [0, l] \rightarrow M$ on a Riemannian manifold M .

Proposition 7.1. *Assumption Let $\epsilon > 0$ and $C : (-\epsilon, \epsilon) \times [0, l] \rightarrow M$ be a C^∞ map. Set*

$$c_{s(t)} = c^t(s) := c(s, t), \quad (s, t) \in (-\epsilon, \epsilon) \times [0, l],$$

Assume $c_0(t)$ has constant speed at $s = 0$, $\|c'_0(t)\| = \|c'_0(0)\|$, $t \in [0, l]$. Define the variation vector field $X(t)$ and tangent field $T(s, t)$ by

$$\begin{aligned} X(t) &:= (c^t)'(0), \quad t \in [0, l] \\ T(s, t) &:= c'_{s(t)}, \quad (s, t) \in (-\epsilon, \epsilon) \times [0, l]. \end{aligned}$$

Note: By definition, $X(t) = dc(\frac{\partial}{\partial s})|_{s=0}$ $T(s, t) = dc(\frac{\partial}{\partial t})$

Problem 10. Let $c : [0, l] \rightarrow M$ be a C^∞ curve and $X(t)$ a vector field along c . Show $\exists$ a variation $\{c_s\}_{s \in (-\epsilon, \epsilon)}$ such that $c_0 = c$ with variation vector field $X(t)$. ■

Theorem 7.2 (First Variation Formula). *For a variation $\{c_s\}$ of c_0 with variation vector field X :*

$$\frac{d}{ds}L(c_s)|_{s=0} = \frac{1}{\|c'_0(0)\|} \left([\langle c'_0, C \rangle_{t=0}^{t=l}] - \int_0^l \langle \nabla_{c_0} c'_0, X \rangle dt \right)$$

Proof.

$$\begin{aligned}\frac{d}{ds}L(c_s)|_{s=0} &= \int_0^l \frac{d}{ds} \frac{d}{ds} \langle c'_s(t), c'_s(t) \rangle^{1/2} |_{s=0} dt \\ &= \frac{1}{2\|c'_0(0)\|} \int_0^l \frac{d}{ds} \langle c'_s(t), c'_s(t) \rangle |_{s=0} dt\end{aligned}$$

By (i) $\frac{1}{2} \frac{d}{ds} \langle c'_s, c'_s \rangle = \langle \nabla_{c'_s} c'_s(t), c'_s(t) \rangle |_{s=0} = \langle \nabla_{c^t} T, T \rangle |_{s=0}$ Since $[\frac{\partial}{\partial s}, \frac{\partial}{\partial t}] = 0$, (i) implies

$$\nabla_{c^t} T = \nabla_{\partial/\partial s}^c dc(\frac{\partial}{\partial s}) = \nabla_{c_0} X$$

Using (3) again,

$$\begin{aligned}\langle \nabla_{c_0} X, T \rangle &= \frac{d}{dt} \langle T, X \rangle |_{s=0} - \langle \nabla_{c_0} T, X \rangle \\ &= \frac{d}{dt} \langle c'_0(t), X \rangle - \langle \nabla_{c_0} c'_0, X \rangle\end{aligned}$$

Thus

$$\begin{aligned}\frac{d}{ds}L(c_s)|_{s=0} &= \frac{1}{\|c'_0(0)\|} \int_0^l \left(\frac{d}{dt} \langle c'_0(t), X \rangle - \langle \nabla_{c_0} c'_0, X \rangle \right) dt \\ &= \frac{1}{\|c'_0(0)\|} \left([c'_0, X]_{t=0}^{t=l} - \int_0^l \langle \nabla_{c_0} c'_0, X \rangle dt \right)\end{aligned}$$

□

Proposition 7.3. *If c is a geodesic, it has constant speed, in other word $\|c_{c(t)}\|$ is independent of t .*

Proof. Apply Lemma 6.7 with $X(t) = Y(t) = c'(t)$ □

Theorem 7.4. *If a constant speed C^∞ curve $c : [0, l] \rightarrow M$ is the shortest among all C^∞ curves connecting $p = c(0)$ and $g = c(l)$, then c is a geodesic.*

Proof. Assume that c is the shortest curve but not a geodesic. Then $\exists t_0 \in (0, l)$ such that $(\nabla_c c')(t_0) \neq 0$. Take a local coordinate. nhd (v, ϕ) such that $c(t_0) \in U$. By continuity, $\exists \delta > 0$ such that $(\nabla_c c')(t) \neq 0$ for $|t - t_0| < \delta$. Let $f : [0, l] \rightarrow \mathbb{R}$ be a C^∞ -function such that $f(t) > 0$ for $|t - t_0| < \delta$ and $f(t) = 0$ otherwise. By Problem 10, $\exists$ variation $\{c_s\}$ with variation vector field $X(t) := f(t) \cdot \nabla_c c'(t)$. Then $\langle (\nabla_c c')(t), X(t) \rangle = f(t) \|\nabla_c c'(t)\|^2$ which is positive for $|t - t_0| < \delta$ and zero otherwise. By the First Variation Formula:

$$\frac{d}{dt}L(c_s)|_{t=0} = -\frac{1}{\|c'(0)\|} \int_{t_0-\delta}^{t_0+\delta} f(t) \|\nabla_c c'(t)\|^2 dt < 0$$

This contradicts the minimality of $L(c_0)$ □

For a coordinate chart (U, ϕ) of M , Let $\phi(c(t)) = (c^1(t), \dots, c^n(t))$. Then $\nabla_c c' = 0$ is:

$$\frac{d^2 c^k}{dt^2} + \sum_{ij=1}^n \Gamma_{ij}^k \frac{dc^i}{dt} \cdot \frac{dc^j}{dt} = 0, \quad k = 1, \dots, n$$

Rewriting with $X^k = \frac{dc^k}{dt}$, we have

$$\frac{dX^k}{dt} + \sum_{ij=1}^n \Gamma_{ij}^k X^i(t) X^j(t) = 0$$

This is a first order nonlinear ordinary differential equation.

Proposition 7.5. $\forall p \in M$ and $\forall v \in T_p M$, $\exists \delta > 0$ and a unique geodesic $c : (-\delta, \delta) \rightarrow M$ such that $c(0) = p$ and $c'(0) = v$.

Proposition 7.6. Let $a < 0 < b$. If two geodesics $\sigma, \tau : (a, b) \rightarrow M$ satisfy $\sigma(0) = \tau(0)$ and $\sigma'(0) = \tau'(0)$, then $\sigma(t) = \tau(t) \quad \forall t \in (a, b)$.

Proof. Show $\sigma(t) = \tau(t)$ for $t \in [0, b)$. Let

$$t_0 : \sup\{s \in [0, b) \mid \sigma(t) = \tau(t) \forall t \in [0, s)\}$$

By Proposition 7.5, $t_0 > 0$. Assume $t_0 < b$. Let $t_i := t_0 - \frac{1}{i}$ ($i \gg 0$) Since σ, τ are continuous and $\sigma(t_i) = \tau(t_i)$, then $\sigma(t_0) = \tau(t_0)$. $\sigma = \tau$ on $[0, t_0] \implies \sigma'(t_0) = \tau'(t_0)$ By Proposition 7.5, $\sigma = \tau$ in a nhd of t_0 , contradicting the definition of t_0 $\square$

Definition 7.7. A geodesic $c(t)$ extended to its maximal possible domain in both directions is a maximal geodesic. $\blacksquare$

For $p \in M$, $v \in T_p M$, let $\gamma_v : J_v \rightarrow M$ be the maximal geodesic such that $\gamma_v(0) = p$, $\gamma_v'(0) = v$. Note:

- (i) $J_v \subset \mathbb{R}$ is an open interval containing 0.
- (ii) $\forall a, t \in \mathbb{R}, \forall v \in TM$
 $t \in J_{av} \equiv at \in J_v$ and $\gamma_{av}(t) = \gamma_v(at)$
- (iii) $\|\gamma_v'(t)\| = \text{const} \implies$ solution of the geodesic equation do not explode within coordinate charts and extend to the boundary.

Proposition 7.8. For $\{t_i\}_{i=1}^\infty \subset J_v$, if $t_i \rightarrow \sup J_v$ (or $\inf J_v$), then $\{\gamma_v(t_i)\}_{i=1}^\infty$ does not converge in M .

Proposition 7.9. $\forall p \in M, \quad v \in T_p M, \quad t_0 \in J_v, \exists O \subset TM : \text{neighbourhood of } v$ such that $\forall w \in O, t_0 \in J_w$. Moreover,

$$\begin{aligned} TM &\rightarrow M \text{ is } C^\infty \\ v &\rightarrow \gamma_v(t_0) \end{aligned}$$

Problem 11. Let $(\bar{M}, \bar{g})$ be a Riemannian manifold, $M \subset \bar{M}$ a submanifold, and $\bar{\nabla}, \nabla$ be the respective Levi-Cevita connections. $\forall p \in M$, let $\pi : T_p \bar{M} \rightarrow T_p M$ be the orthogonal projection. **Show:** $(\nabla_X Y)_p = \pi((\bar{\nabla}_X Y)_p)$, $X, Y \in \mathbb{X}(M)$ ■

Example 7.10. Find the geodesic of γ with arc-length parameter in the 2-dimensional unit sphere $\mathbb{S}^2 \subset \mathbb{R}^3$. By Problem 11.

$$\nabla_\gamma \gamma' = \pi(\gamma''(t)) = 0,$$

here, $\pi : \mathbb{R}^3 \rightarrow T_p \mathbb{S}^2$, ($p \in \mathbb{S}^2$) is the orthogonal projection. This implies that $\gamma''(t)$ is parallel to the line segment connecting O with $\gamma(t)$. Thus $\exists f(t)$ such that $\gamma''(t) = f(t) \cdot \gamma(t)$. Differentiating $\|\gamma(t)\|_{\mathbb{R}^3}^2 = 1$ twice,

$$\begin{aligned} 0 &= \frac{d}{dt} \langle \gamma(t), \gamma(t) \rangle_{\mathbb{R}^3} = 2 \langle \gamma'(t), \gamma(t) \rangle_{\mathbb{R}^3} \\ f(t) &= \langle \gamma''(t), \gamma(t) \rangle_{\mathbb{R}^3} = -\langle \gamma'(t), \gamma'(t) \rangle_{\mathbb{R}^3} = -1 \end{aligned}$$

We obtain $\gamma''(t) + \gamma(t) = 0$.

$\gamma(t) = \cos(t)\gamma(0) + \sin(t)\gamma'(0)$ which represents a great circle. ■

Problem 12. Let $H := \{(x, y) \in \mathbb{R}^2 | y > 0\}$ be the upper half-plane. The hyperbolic plane (H, g) is defined by the metric $g = \frac{dx^2 + dy^2}{y^2}$. Show that the geodesic equation are:

$$\begin{cases} x''(t) - \frac{2x'(t)y'(t)}{y(t)} = 0 \\ y''(t) + \frac{(x'(t))^2 - (y'(t))^2}{y(t)} = 0 \end{cases}$$

(Note: Assuming unitt speed $\frac{(x'(t))^2 + (y'(t))^2}{y(t)^2} = 1$, the geodesic (H, g) are vertical half-lines or semicircles centered on the x -axis.) ■

8 Curvature tensor

Let M be a Riemannian manifold. $\forall p \in M, \forall u, v \in T_p M$ choose a C^∞ map

$$c : (-\epsilon, \epsilon) \times (-\delta, \delta) \rightarrow M$$

s.t. $u = (c^0)'(0, 0)$, $v = (c^1)'(0, 0)$, $p = c(0, 0)$, where $c_s(t) = c^t(s) = c(s, t)$. Define parallel transport around the square:

$$P_{s,t}w := P_{c^0(0)}P_{c_s(0)}P_{c^t(s)}P_{c_o(t)}w, \quad w \in T_p M.$$

In Euclidean space, $P_{s,t}w = \text{Id } w \forall s, t$. In general $P_{s,t}w - w \neq 0$.

Lemma 8.1. For $p \in M$ and $w \in T_p M$, the map

$$\begin{aligned} (-\epsilon, \epsilon) \times (-\delta, \delta) &\rightarrow T_p M \\ (s, t) &\mapsto P_{s,t}w \end{aligned}$$

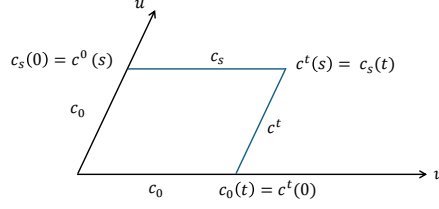


Figure 1: Parallel transport around a parallelogram.

is C^∞ and satisfies

$$\left. \frac{\partial^2}{\partial s \partial t} P_{s,t} w \right|_{(s,t)=(0,0)} = \lim_{(s,t) \rightarrow (0,0)} \frac{P_{s,t} w - w}{st},$$

where the left-hand side is the Fréchet derivative for vector-valued functions. Specifically, for an orthonormal basis $e_1, \dots, e_n$ of $T_p M$,

$$\frac{\partial^2}{\partial s \partial t} P_{s,t} w = \sum_{i=1}^n \left(\frac{\partial^2}{\partial s \partial t} \langle P_{s,t} w, e_i \rangle \right) e_i.$$

Proof. Consider s, t sufficiently small. Assume $\mathfrak{S}(c)$ is contained in a coordinate neighborhood (U, ϕ) . First, consider $P_{c_0(t)} w$. Let,

$$\phi(c_0(t)) = (c_0^1(t), \dots, c_0^n(t)), \quad P_{c_0(t)} w = \sum_{i=1}^n \xi^i(t) (\partial_i)_{c_0(t)},$$

then $\xi^1(t), \dots, \xi^n(t)$ solve the linear ODE

$$\frac{d\xi^k}{dt} + \sum_{i,j=1}^n \Gamma_{ij}^k \frac{dc_0^i}{dt} \xi^j = 0, \quad k = 1, \dots, n,$$

with initial condition $\sum_{i=1}^n \xi^i(0) (\partial_i)_p = w$. Next, consider $P_{c^t(s)} P_{c_0(t)} w$. Let

$$\sum_{i=1}^n \eta_t^i(s) (\partial_i)_{c^t(s)} = P_{c^t(s)} P_{c_0(t)} w.$$

Then $\eta_t^1(s), \dots, \eta_t^n(s)$ solve the linear ODE

$$\frac{d\eta_t^k}{ds} + \sum_{i,j=1}^n \Gamma_{ij}^k \frac{d(c^t)^i}{ds} \eta_t^j = 0, \quad k = 1, \dots, n,$$

with initial conditions $\eta_t^k(0) = \xi^k(t)$ ($k = 1, \dots, n$). By standard ODE theory, $\eta_t^k(s)$ are C^∞ functions in (s, t) . Repeating the same argument shows $P_{s,t} w$ is C^∞ in (s, t) .

Fix an orthonormal basis $e_1, \dots, e_n$ of $T_p M$ and Taylor expand $\langle P_{s,t} w, e_i \rangle$ at $(s, t) = (0, 0)$. Since $P_{s,0} = P_{0,t} w = w$,

$$\frac{\partial^k}{\partial s^k} \langle P_{s,0}, w, e_i \rangle = \frac{\partial^k}{\partial t^k} \langle P_{0,t} w, e_i \rangle = 0, \quad k \geq 1.$$

Then, $\forall i$,

$$\langle P_{s,t} w, e_i \rangle = \langle w, e_i \rangle + st \frac{\partial^2}{\partial s \partial t} \langle P_{s,t} w, e_i \rangle \Big|_{(0,0)} + o(st^2 + s^2 t).$$

It follows that

$$\left\langle \lim_{(s,t) \rightarrow (0,0)} \frac{P_{s,t} w - w}{st}, e_i \right\rangle = \frac{\partial^2}{\partial s \partial t} \langle P_{s,t} w, e_i \rangle \Big|_{(s,t)=(0,0)}.$$

Multiplying by e_i and summing over i gives the desired result. $\square$

Definition 8.2 (Curvature tensor). $\forall p \in M$, $u, v \in T_p M$, define $R(u, v) : T_p M \rightarrow T_p M$ by

$$R(u, v)w := \frac{\partial^2}{\partial s \partial t} P_{s,t} w \Big|_{(s,t)=(0,0)} = \lim_{(s,t) \rightarrow (0,0)} \frac{P_{s,t} w - w}{st}, \quad w \in T_p M.$$

The map $R : T_p M \times T_p M \times T_p M \rightarrow T_p M$ is called the *curvature tensor*. R is a tensor field of type $(1, 3)$. $\blacksquare$

Lemma 8.3. $\forall p \in M$ and $\forall w \in T_p M$, let $Z_{(s,t)} := P_{c^t(s)} P_{c_o(t)} w$. Then

$$R(u, v)w = (\nabla_c + \nabla_{c_s} Z) \Big|_{(s,t)=(0,0)}.$$

Proof. By Proposition 6.11, the covariant derivative along c_s is

$$(\nabla_{c_s} Z) \Big|_{t=0} = \lim_{h \rightarrow 0} \frac{P_{c_s(0)} Z(s, h) - Z(s, 0)}{h}.$$

Note that $c_0(h) = c^h(0)$ and $P_{c^h(0)} : T_{c^h(0)} M \rightarrow T_{c^h(0)} M$ is the identity map. Since $P_{c_0(h)} : T_p M \rightarrow T_{c_0(h)}$ and $P_{c_0(0)} : T_{c_0(h)} \rightarrow T_p M$ are inverse to each other. We have

$$\begin{aligned} P_{c_0(0)} Z(s, h) &= P_{c_0(0)} P_{c^h(0)} P_{c_0(h)} w \\ &= P_{c_0(0)} P_{c_0(h)} w = w = Z(0, 0). \end{aligned}$$

Consequently, $(\nabla_{c_s} Z) \Big|_{(s,t)=(0,0)} = (\nabla_{c_0} Z) \Big|_{t=0} = 0$. Thus, the second covariant

derivative satisfies

$$\begin{aligned}
(\nabla_c + \nabla_{c^s} Z)|_{(s,t)=(0,0)} &= (\nabla_{c^0} \nabla_{c^s} Z)|_{s=0} \\
&= \lim_{k \rightarrow 0} \frac{P_{c^0(0)}((\nabla_{c^s} Z)|_{(s,t)=(k,0)}) - (\nabla_{c^s} Z)|_{(s,t)=(0,0)}}{k} \\
&= \lim_{k \rightarrow 0} \frac{P_{c^0(0)} \left(\lim_{h \rightarrow 0} \frac{P_{c_k(0)} Z(k,h) - Z(k,0)}{h} \right)}{k} \\
&= \lim_{k \rightarrow 0} \lim_{h \rightarrow 0} \frac{P_{c^0(0)} \left(\frac{P_{c_k(0)} P_{c^h(k)} P_{c_0(h)} w - P_{c^0(k)} w}{h} \right)}{k} \\
&= \lim_{k \rightarrow 0} \lim_{h \rightarrow 0} \frac{P_{k,h} w - w}{hk} = R'.
\end{aligned}$$

□

Definition 8.4 (Curvature tensor in coordinates). On a coordinate chart of M , define

$$R_{ijk}^l := \partial_i \Gamma_{jk}^l - \partial_j \Gamma_{ik}^l + \sum_{m=1}^n (\Gamma_{im}^l \Gamma_{jk}^m - \Gamma_{jm}^l \Gamma_{ik}^m)$$

with $i, j, k, l = 1, \dots, n$. ■

Theorem 8.5. $\forall p \in M, u, v, w \in T_p M$, let $u = \sum_{i=1}^n u^i (\partial_i)_p$, $v = \sum_{j=1}^n v^j (\partial_j)_p$ and $w = \sum_{k=1}^n w^k (\partial_k)_p$ in a chart. Then

$$R(u, v)w = \sum_{i,j,k,l=1}^n u^i v^j w^k R_{ijk}^l (\partial_l)_p,$$

multilinear in u, v, w . In particular, $R(u, v)w$ is independent of the choice of the chart.

Proof. Using notation from Lemma 8.3, Z is parallel along c^t . Thus,

$$\nabla_{\partial/\partial s}^c Z = 0 \Rightarrow \nabla_{\partial/\partial t}^c \nabla_{\partial/\partial s}^c Z = 0.$$

Since $[\frac{\partial}{\partial s}, \frac{\partial}{\partial t}]$, we have

$$\begin{aligned}
&\left(\nabla_{\partial/\partial s}^c \nabla_{\partial/\partial t}^c Z - \nabla_{\partial/\partial t}^c \nabla_{\partial/\partial s}^c Z \right) \Big|_{(0,0)} \\
&= \left(\nabla_{\partial/\partial s}^c \nabla_{\partial/\partial t}^c Z \right) \Big|_{(0,0)} = (\nabla_{c^t} \nabla_{c^s} Z)|_{(0,0)} = R(u, v)w.
\end{aligned}$$

The result follows by combining this with Problem 13. □

Problem 13. Let N be a manifold, $\xi : N \rightarrow M$ a C^∞ map; X, Y vector fields on N and Z a vector field along ξ . In a coordinate chart of M , where

$$d\xi(X) = \sum X^i \partial_i, \quad d\xi(Y) = \sum Y^j \partial_j, \quad Z = \sum Z^k \partial_k.$$

Show that

$$\nabla_X^\xi \nabla_Y^\xi Z - \nabla_Y^\xi \nabla_X^\xi Z = \sum_{i,j,k,l=1}^n X^i Y^j Z^k R_{ijk}^l \partial_l.$$

■

The following follows from Problem 13.

Corollary 8.6. (1) For ξ, X, Y, Z as in Problem 13,

$$\nabla_X^\xi \nabla_Y^\xi Z - \nabla_Y^\xi \nabla_X^\xi Z - \nabla_{[X,Y]}^\xi Z = R(d\xi(X), d\xi(Y))Z.$$

(2) For vector fields X, Y, Z on M

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z.$$

Let $R_{ijkl} := \langle R(\partial_i, \partial_j)\partial_k, \partial_l \rangle$ be the coefficients of the $(0, 4)$ -tensor field. By Theorem 8.5,

$$R_{ijkl} = \sum_{m=1}^n R_{ijk}^m g_{ml}, \quad i, j, k, l = 1, \dots, n.$$

The fundamental symmetries of the curvature tensor are given in Problem 14.

Problem 14. Show that $\forall p \in M, \quad u, v, w, x \in P_p M$ and indices i, j, k, l :

- (1) $R(u, v)w = -R(v, u)w, \quad R_{jik}^l = -R_{ijk}^l.$
- (2) $R(u, v) + R(v, w) + R(w, u) = 0.$

$$R_{ijk}^l + R_{jki}^l + R_{kij}^l = 0 \text{ (first Bianchi identity).}$$

- (3) $\langle R(u, v)x, w \rangle = -\langle R(v, u)w, x \rangle, \quad R_{ijkl} = -R_{jikl}.$
- (4) $\langle R(u, v)w, x \rangle = \langle R(x, u)v, w \rangle, \quad R_{ijkl} = R_{klij}.$

■

9 Sectional and Ricci curvatures

Let M be a Riemannian manifold.

Lemma 9.1. $\forall p \in M$, let $\sigma \subset T_p M$ be a 2-dimensional subspace. For any basis $\{u, v\}$ of σ ,

$$K_\sigma := \frac{\langle R(u, v)u, v \rangle}{\|u \times v\|^2}$$

is independent of $\{u, v\}$, where

$$\|u \times v\| := \sqrt{\|u\|^2 \|v\|^2 - \langle u, v \rangle^2}$$

is the area of the parallelogram spanned by u, v .

Proof. Fix an orthonormal basis $\{e_1, e_2\}$ of $\sigma \forall \{u, v\}$ basis of σ , let

$$u = u^1 e_1 + u^2 e_2, \quad v = v^1 e_1 + v^2 e_2.$$

Then

$$\begin{aligned} \langle R(u, v)v, u \rangle &= \sum_{i,j,k,l=1,2} u^i v^j u^k u^l \langle R(e_i, e_j)e_k, e_l \rangle \\ &= [(u^1 v^2)^2 + (u^2 v^1)^2 - 2u^1 u^2 v^1 v^2] \langle R(e_1, e_2)e_2, e_1 \rangle \\ \Rightarrow \frac{\langle R(u, v)v, u \rangle}{\|u \times v\|^2} &= \langle R(e_1, e_2)e_2, e_1 \rangle. \end{aligned}$$

□

Definition 9.2 (Sectional curvature). $\forall o \in M, \forall \sigma \subset T_p M$ a 2-dimensional subspace and K_σ defined as in Lemma 9.1 is the *sectional curvature*. ■

Remark 9.3. The sectional curvature is undefined if $\dim M = 1$. If $\dim M = 2$, $\sigma = T_p M \Rightarrow K_\sigma$ is a function on M . ■

Theorem 9.4 (Sectional curvatures determine the Riemann curvature tensor). *At each $p \in M$, the sectional curvatures*

$$\{K_\sigma | \sigma \subset T_p M \text{ is 2-dimensional}\}$$

determine the Riemann curvature tensor R at p .

Proof. $\forall u, v, w \in T_p M$, define

$$K(u, v) := \langle R(u, v)v, u \rangle, \quad R_w(u, v) := \langle R(u, w)w, v \rangle.$$

If $\{u, v\}$ is linearly independent $\Rightarrow K(u, v) = K_\sigma \|u \times v\|^2$ for $\sigma = \text{Span}_{\mathbb{R}}\{u, v\}$.
If $\{u, v\}$ is linearly dependent $\Rightarrow K(u, v) = 0$ by Problem 14. Thus, sectional curvatures determine K .

F_w is a symmetric bilinear form

$$\begin{aligned} F_w(u, v) &= \langle R(u, w)w, v \rangle \\ &= \langle R(w, v)u, w \rangle && \text{Problem 14 (4)} \\ &= -\langle R(v, w)u, w \rangle && \text{Problem 14 (1)} \\ &= \langle R(v, w)w, u \rangle && \text{Problem 14 (3)} \\ &= F_w(v, w). \end{aligned}$$

Polarization yields

$$\begin{aligned} 2F_w(u, v) &= F_w(u + v, u + v) - F_w(u, u) - F_w(v, v) \\ &= K(u + v, w) - K(u, w) - K(v, w). \end{aligned}$$

K determines F_W . Expanding R yields

$$\begin{aligned} R(u, v+w)(v+w) &= R(u, v)v + R(u, v)w + R(u, w)v + R(u, w)w \\ R(u+w, v)(u+w) &= R(u, v)u + R(u, v)w + R(w, u)u + R(w, v)w \\ &= R(u, v)w + R(v, u)w \end{aligned}$$

$$R(u, v+w) - R(v, u+w)(u+w) = 3R(u, v)w + R(u, v)v - R(v, u)u + R(u, w)w - R(v, w)w.$$

Taking the inner product with $x \in T_p M$ and solving for $\langle R(u, v)w, x \rangle$:

$$\begin{aligned} \langle R(u, v)w, x \rangle &= \frac{1}{3} \{ F_{v+w}(u, x) + F_{u+w}(v, x) \\ &\quad F_v(u, x) + F_u(u, x) \\ &\quad - F_w(u, x) - F_w(v, x) \}. \end{aligned}$$

Thus, R is uniquely determined by K_σ . □

Corollary 9.5. $\forall p \in M$, the following are equivalent:

- (1) $R(u, v)w = 0 \quad \forall u, v, w \in T_p M$.
- (2) $K_\sigma = 0 \quad \forall$ 2-dimensional $\sigma \subset T_p M$.

Definition 9.6 (Trace of a bilinear form). Let V be an n -dimensional inner product space and $f : V \times V \rightarrow \mathbb{R}$ a bilinear form. The *trace* of f is

$$\text{tr } f := \sum_{i=1}^n f(e_i, e_i)$$

where $\{e_1, \dots, e_n\}$ is any orthonormal basis of V . ■

Remark 9.7. Well-definedness (independence of $\{e_i\}_{i=1}^n$) is left as an exercise to the reader. ■

Definition 9.8 (Ricci and scalar curvature). $\forall p \in M, \forall v, w \in T_p M$, $\text{Ric}(v, w)$ is the trace of the bilinear form

$$\begin{aligned} T_p M \times T_p M &\rightarrow \mathbb{R}, \\ (x, y) &\mapsto \langle R(x, v)w, y \rangle. \end{aligned}$$

For an orthonormal basis $\{e_1, \dots, e_n\}$ of $T_p M$,

$$\text{Ric}(v, w) = \sum_{i=1}^n \langle R(e_i, v)w, e_i \rangle.$$

The symmetric $(0, 2)$ -tensor Ric is the *Ricci curvature tensor*.

For $v \in T_p M$ with $\|v\| = 1$, $\text{Ric}(v, v)$ is the *Ricci curvature* in direction v . The trace of the Ricci tensor,

$$\text{Scal}(p) := \sum_{i=1}^n \text{Ric}(e_i, e_i),$$

is the *scalar curvature* at p . ■

Proposition 9.9. $\forall v \in T_p M$ with $\|v\| = 1$, let $\{e_1, \dots, e_n = v\}$ be an orthonormal basis of $T_p M$. Furthermore, let $\sigma_i \subset T_p M$ ($i = 1, \dots, n-1$) be the 2-dimensional subspace spanned by $\{e_i, v\}$ for $i = 1, \dots, n-1$. Then,

$$\text{Ric}(v, v) = \sum_{i=1}^{n-1} K_{\sigma_i}.$$

Remark 9.10. If $\exists c \in \mathbb{R}$ such that $K_{\sigma} \geq c \forall \sigma$ 2-dimensional subspace $\subset T_p M$ 2-dimensional $\Rightarrow \text{Ric}(v, v) \geq (n-1)c \forall$ unit vector $v \in T_p M$ (with $\|v\| = 1$). This is conventionally denoted as $\text{Ric} \geq (n-1)c$. ■